

MCLF - the book

Models of curves over local fields

Edited by Stefan Wewers

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Preface

The MCLF project (*Models of Curves over Local Fields*) develops a SageMath toolbox for computations with models of curves over local fields, with particular emphasis on semistable reduction. The project is hosted at <https://github.com/MCLF/mclf>.

This book grew out of an Oberseminar held during the winter semester 2024/25 at the Institute of Algebra and Number Theory at Ulm University. The seminar was organized as a short series of talks under the title *Models of curves*. Its original aim was to develop, from both a theoretical and a computational perspective, some of the basic ideas underlying the MCLF project.

The seminar consisted of five talks:

- Miriam Ní Chobhtaigh: *Models of curves and valuations*;
- Max Schwegele: *Proof of Theorem 1*;
- Xiaodong Zhang: *Models with reduced special fiber*;
- Robert Nowak: *The Epp problem for p -cyclic covers of curves*;
- Kletus Stern: *Mac Lane valuations*.

The initial plan was simply to revise and combine the notes prepared for these talks into a larger set of seminar notes. As work on the project progressed, however, it became clear that the material would benefit from a broader and more systematic treatment. In particular, the relation between algebraic models, valuations, Berkovich geometry, Mac Lane theory, and explicit computations suggested a structure extending well beyond the original five talks.

The present book is the result of this development. The original seminar notes were compiled, edited and extended by Maximilian Fülle. The content of the first four talks is the germ of Part I, *Models and valuations*. The notes for the fifth talk form the starting point of Part II, *Analytic and computational tools*.

Julian R  th's thesis, *Models of curves and valuations* [R  14], has been an important foundation for both the MCLF project and this book. It brings together the description of normal models by valuations, Mac Lane theory, and computational

methods in a form that has strongly influenced the organization and mathematical approach of the project. We gratefully acknowledge the importance of this work.

The book is a collaborative and evolving project. Individual chapters identify their authors, while the volume as a whole is edited by Stefan Wewers. The chapter attributions refer to the finished chapters, not merely to the speakers in the original seminar: in some cases the seminar notes have been substantially revised, reorganized, or supplemented by additional authors.

We thank all participants in the Oberseminar for their talks, notes, questions, and discussions. Without the seminar and the work invested in its preparation, this book would not have been started.

The preparation of this book has been assisted by generative-AI tools and AI agents. They have been used mainly for editorial and technical tasks, including revising and reorganizing prose, adapting material from existing notes and sources, checking consistency, editing $\text{\LaTeX}$, and maintaining the repository. In some cases, they have also helped to expand routine steps in arguments whose mathematical content and strategy had already been supplied by the authors. All mathematical claims, proofs, examples, computations, and references remain the responsibility of the human authors and editors, who have reviewed and approved the text. AI systems are not regarded as authors or as independent mathematical contributors.

Part I

Models and valuations

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This part develops the algebraic foundations and valuation theory used throughout the book. It introduces models of curves, attaches geometric valuations to components of their special fibres, establishes the correspondence between models and finite sets of these valuations, and studies their behaviour under finite extensions of the base field.

Chapter 1

Models of curves over valued fields

This chapter introduces models of smooth projective curves over valued fields, together with their morphisms and the resulting partial order by domination. After discussing normality and basic examples, we attach a geometric valuation and a multiplicity to every irreducible component of the special fibre; the resulting finite set of valuations is the starting point for the correspondence developed in Chapter 2.

1.1 Valued fields, curves, and models

We begin by fixing the valued-field notation and standing hypotheses used throughout Part I, and then introduce the basic notion of a model of a curve.

Let K be a field endowed with an additive valuation of rank 1, $v_K : K \rightarrow \hat{\mathbb{R}} := \mathbb{R} \cup \{\infty\}$. Let $\Gamma_K := v_K(K^*) \subset \mathbb{R}$ be the value group of v_K , $\mathcal{O}_K \subset K$ its valuation ring with maximal ideal $\mathfrak{m}_K$ and $k := \mathcal{O}_K/\mathfrak{m}_K$ its residue field.

Unless explicitly stated otherwise, the following assumption remains in force throughout Part I. We indicate where the individual hypotheses are used.

Assumption 1.1. (a) v_K is discrete, i.e. Γ_K is a discrete subgroup of $\mathbb{R}$,

(b) K is complete with respect to v_K ,

(c) k is perfect.

Throughout the monograph, X will denote a *nice* curve over K , that is a smooth, projective and absolutely irreducible curve over K .

Definition 1.2. A *model* $\mathcal{X}$ of X is a proper, flat and normal $\mathcal{O}_K$ -scheme with an identification of its generic fibre $\mathcal{X}_\eta$ with X .

Example 1.3. Let $K = \mathbb{Q}_p$, $\mathcal{O}_K = \mathbb{Z}_p$ and $X = \mathbb{P}_K^1$. Then $\mathcal{X} = \mathbb{P}_{\mathcal{O}_K}^1$ is a model of X with the evident identification $\mathcal{X}_\eta = \mathcal{X}_K \cong \mathbb{P}_K^1$. The special fibre $\mathcal{X}_s$ has one irreducible component which is isomorphic to $\mathbb{P}_k^1$.

Remark 1.4. The condition of a model being flat is equivalent to it being integral: if $\mathcal{X}$ is integral and its structural map is compatible with the given generic fibre, then every local ring $\mathcal{O}_{\mathcal{X},x}$ is a domain containing $\mathcal{O}_K$. Thus $\mathcal{O}_{\mathcal{X},x}$ is a torsion-free module over the discrete valuation ring $\mathcal{O}_K$, hence flat by [Liu02, Corollary 1.2.14]. Conversely, the integrality of a model $\mathcal{X}$ of X follows from its flatness and the integrality of X by [Liu02, Proposition 4.3.8].

Proposition 1.5. *Every nice curve X/K admits a model.*

Proof. Choose a closed immersion $X \hookrightarrow \mathbb{P}_K^n$, and let $\mathcal{X}_0$ be the schematic closure of X in $\mathbb{P}_{\mathcal{O}_K}^n$. Then $\mathcal{X}_0$ is proper and integral over $\mathcal{O}_K$, and hence flat by Remark 1.4. Normalize $\mathcal{X}_0$ in F_X . The completeness hypothesis in Assumption 1.1 implies that the discrete valuation ring $\mathcal{O}_K$ is excellent, so this normalization is finite by [Liu02, Theorem 8.2.39(d)]. It is therefore proper over $\mathcal{O}_K$; it is normal and integral by construction, hence flat, and its generic fibre is X because X is normal. Thus the normalization is a model of X . $\square$

1.2 The category of models and modifications

We now introduce the morphisms used to compare models of a fixed curve. They turn the collection of models into a partially ordered system in which any two models admit a common dominating model.

Definition 1.6. Let $\mathcal{X}$ and $\mathcal{X}'$ be models of X . A *modification* $\varphi : \mathcal{X}' \rightarrow \mathcal{X}$ is a morphism of $\mathcal{O}_K$ -schemes whose restriction to the generic fibres is the identity on X under the fixed identifications $\mathcal{X}'_\eta \cong X \cong \mathcal{X}_\eta$. In this situation, we say that $\mathcal{X}'$ *dominates* $\mathcal{X}$.

Lemma 1.7. *Between two fixed models of X , there is at most one modification.*

Proof. Let $\varphi, \psi : \mathcal{X}' \rightarrow \mathcal{X}$ be modifications. They agree on the generic fibre $\mathcal{X}'_\eta$, which is dense in the integral scheme $\mathcal{X}'$. Since $\mathcal{X}$ is separated, the equalizer of φ and ψ is closed in $\mathcal{X}'$. It contains the dense generic fibre and is therefore all of $\mathcal{X}'$, so $\varphi = \psi$. $\square$

Proposition 1.8. *Let $\varphi : \mathcal{X}' \rightarrow \mathcal{X}$ be a modification of models. Then φ is proper, dominant, and surjective. Moreover, there is a finite set $S \subset \mathcal{X}_s$ of closed points such that φ is an isomorphism over $\mathcal{X} \setminus S$ and, for every $x \in S$, the fibre $\varphi^{-1}(x)$ is a connected projective curve over $\kappa(x)$.*

Proof. Let $\varphi : \mathcal{X}' \rightarrow \mathcal{X}$ be a modification. Since $\mathcal{X}$ is separated over $\text{Spec } \mathcal{O}_K$, the graph morphism

$$\Gamma_\varphi : \mathcal{X}' \longrightarrow \mathcal{X}' \times_{\text{Spec } \mathcal{O}_K} \mathcal{X}$$

is a closed immersion. The projection $\mathcal{X}' \times_{\text{Spec } \mathcal{O}_K} \mathcal{X} \rightarrow \mathcal{X}$ is proper because it is the base change of the proper morphism $\mathcal{X}' \rightarrow \text{Spec } \mathcal{O}_K$. Their composite is φ , so

φ is proper. Its image contains the generic fibre X , which is dense in $\mathcal{X}$, and hence φ is dominant. Finally, a proper morphism has closed image, so the image of φ is both dense and closed in $\mathcal{X}$ and must equal $\mathcal{X}$.

A proper birational morphism to a normal scheme is an isomorphism at every codimension-one point of the target; this follows by applying the valuative criterion of properness to the corresponding discrete valuation ring. Hence the complement of the maximal open subset of $\mathcal{X}$ over which φ is an isomorphism is a closed subset of codimension at least two. Since $\mathcal{X}$ is a two-dimensional Noetherian scheme and φ is already an isomorphism on the generic fibre, this complement is a finite set $S \subset \mathcal{X}_s$ of closed points.

By the connectedness form of Zariski's Main Theorem, the fibres of φ are connected; see [Aut26, Lemma 0AY8]. If $x \in S$, the fibre over x cannot be zero-dimensional: otherwise φ would be finite in a neighbourhood of x , and a finite birational morphism to the normal scheme $\mathcal{X}$ would be an isomorphism there. Thus $\varphi^{-1}(x)$ is a proper one-dimensional scheme over $\kappa(x)$, hence a projective curve. $\square$

We write

$$\mathcal{X} \leq \mathcal{X}'$$

if there is a modification $\mathcal{X}' \rightarrow \mathcal{X}$; thus the larger model is the one that dominates the smaller model. Here and below, models that are isomorphic as models of X are identified when discussing this relation.

Proposition 1.9. *The relation $\leq$ is a partial order on the set of isomorphism classes of models of X . The category of models and modifications is equivalent to the partially ordered set regarded as a category.*

Proof. The identity morphism gives reflexivity, and composition of modifications gives transitivity. If $\mathcal{X} \leq \mathcal{X}'$ and $\mathcal{X}' \leq \mathcal{X}$, there are modifications $\varphi : \mathcal{X}' \rightarrow \mathcal{X}$ and $\psi : \mathcal{X} \rightarrow \mathcal{X}'$. Both composites are modifications and hence, by Lemma 1.7, are the respective identity morphisms. Thus φ and ψ are mutually inverse, which proves antisymmetry on isomorphism classes. The same uniqueness lemma says that there is at most one morphism between any two models, so choosing one representative of each isomorphism class gives the asserted equivalence of categories. $\square$

Proposition 1.10. *Any two models $\mathcal{X}_1$ and $\mathcal{X}_2$ of X admit a common dominating model $\mathcal{X}_3$.*

Proof. Consider the diagonal morphism determined by the fixed identifications of the generic fibres,

$$X \longrightarrow \mathcal{X}_1 \times_{\text{Spec } \mathcal{O}_K} \mathcal{X}_2.$$

Let $\mathcal{Z}$ be the schematic closure of its image. The product $\mathcal{X}_1 \times_{\text{Spec } \mathcal{O}_K} \mathcal{X}_2$ is proper over $\text{Spec } \mathcal{O}_K$, and hence so is its closed subscheme $\mathcal{Z}$. Moreover, $\mathcal{Z}$ is integral and its generic fibre is the diagonal copy of X . It is therefore flat over the discrete valuation ring $\mathcal{O}_K$ by Remark 1.4.

Let $\mathcal{X}_3$ be the normalization of $\mathcal{Z}$ in F_X . The completeness assumption in Assumption 1.1 enters here: it implies that $\mathcal{O}_K$ is excellent, so the normalization is finite; see [Liu02, Chapter 8, in particular Theorem 8.2.39 (d)]. Consequently, $\mathcal{X}_3$ is proper over $\text{Spec } \mathcal{O}_K$. It is integral and hence flat over $\mathcal{O}_K$, and it is normal by construction. Since X is normal, normalization does not change the generic fibre, so $(\mathcal{X}_3)_\eta = X$. Thus $\mathcal{X}_3$ is a model of X . Composing the normalization map with the two projections gives modifications $\mathcal{X}_3 \rightarrow \mathcal{X}_i$ for $i = 1, 2$. $\square$

Remark 1.11. Proposition 1.10 says that the partially ordered set of models is directed upward. Since modifications point from a dominating model to a dominated model, the category of models is correspondingly cofiltered and may be viewed as an inverse system. Common dominating models may also be produced through suitable blowups followed by normalization.

Chapter 2 will identify this order with inclusion between the corresponding finite sets of geometric valuations.

1.3 Local properties of models

We briefly recall some local properties of two-dimensional schemes that will be used throughout the book. Our purpose is only to fix terminology and record the standard implications needed below. For further details, we refer to [Liu02, §4.2, §6.3, §6.4, §8.2.2, and §8.3.1].

For the following discussion, it is sometimes useful to drop the normality assumption of Definition 1.2.

Definition 1.12. A *pre-model* of X is a flat and proper $\mathcal{O}_K$ -scheme $\mathcal{X}$ with $\mathcal{X}_\eta = X$. That is, $\mathcal{X}$ satisfies all the conditions of a model except that it need not be normal.

Under Assumption 1.1, the valuation ring $\mathcal{O}_K$ is an excellent discrete valuation ring. Every pre-model of X is therefore a Noetherian excellent scheme of pure dimension 2. Its local rings at closed points of the special fibre have dimension 2, while the local rings at the generic points of irreducible components of the special fibre have dimension 1. See [Liu02, §8.2.3 and §8.3.1].

Let A be a Noetherian local ring with maximal ideal $\mathfrak{m}$ and residue field κ . The ring A is called regular if

$$\dim A = \dim_\kappa \mathfrak{m}/\mathfrak{m}^2.$$

Equivalently, the maximal ideal of A can be generated by $\dim A$ elements. A scheme is regular at a point if its local ring at that point is regular. We refer to [Liu02, §4.2] for the basic properties of regular local rings and regular schemes.

The ring A is called Cohen–Macaulay if

$$\text{depth } A = \dim A.$$

A Noetherian scheme is Cohen–Macaulay if all its local rings are Cohen–Macaulay. The definition and the main properties of Cohen–Macaulay rings are recalled in [Liu02, §8.2.2].

A Noetherian local ring A is called a *complete intersection* if its completion can be written as the quotient of a regular local ring by an ideal generated by a regular sequence. A locally Noetherian scheme is called *locally a complete intersection* (l.c.i.) if all its local rings are complete intersections. This is a local property and does not assert that the scheme is a complete intersection in a single ambient scheme. The same terminology is often used for morphisms in the literature; we will not need that relative notion here. In particular, an effective Cartier divisor in a regular scheme is locally a complete intersection. See [Liu02, §6.3].

The Gorenstein property is a stronger duality property of Cohen–Macaulay local rings. We will not need its full definition at this point; see [Liu02, §6.4] for the relevant duality theory. For Noetherian local rings, we shall use the standard implications

$$\text{regular} \implies \text{l.c.i.} \implies \text{Gorenstein} \implies \text{Cohen–Macaulay}.$$

None of the converse implications holds in general.

Serre’s criterion says that a Noetherian ring is normal if and only if it satisfies the conditions (R_1) and (S_2) ; see [Liu02, Theorem 8.2.23]. A Cohen–Macaulay ring satisfies (S_2) . Consequently, an integral Cohen–Macaulay scheme is normal if and only if it is regular in codimension 1.

In particular, every model of X is Cohen–Macaulay. Indeed, a model is normal and has dimension 2, so Serre’s condition (S_2) is equivalent to the Cohen–Macaulay condition for its local rings. The same conclusion need not hold for an arbitrary pre-model.

Proposition 1.13. *Let $\mathcal{X}$ be a model of X . Then the nonregular locus of $\mathcal{X}$ is a finite set of closed points contained in the special fibre $\mathcal{X}_s$.*

Proof. This is a standard consequence of normality and excellence. Normality implies that $\mathcal{X}$ is regular in codimension 1, while excellence implies that its regular locus is open. Thus the nonregular locus is a closed subset of codimension at least 2 in the two-dimensional Noetherian scheme $\mathcal{X}$. See [Liu02, §8.2.3 and §8.3.1]. $\square$

Remark 1.14. Regularity of the total space $\mathcal{X}$ should not be confused with regularity, smoothness, or reducedness of its special fibre. A regular model may have a singular special fibre, and its special fibre need not be reduced. Conversely, a model with reduced special fibre need not be regular. These distinctions will be important in the study of permanent and semistable models.

1.4 Normality and examples of models

To recognize when a proper flat scheme is a model, it remains to understand its normality. We first recall a criterion in terms of the special fibre and then apply it

to several examples.

Lemma 1.15. *Let $\mathcal{X}$ be a pre-model of X . Then $\mathcal{X}$ is normal (i.e. a model) if and only if the following two conditions hold.*

- (a) *The special fibre $\mathcal{X}_s$ has no embedded associated points; equivalently, there is no non-generic point $x \in \mathcal{X}_s$ satisfying $\mathfrak{m}_x = \text{Ann}_R(r)$ for some $r \in R := \mathcal{O}_{\mathcal{X},x}$ (see [Liu02, Definition 7.1.6]).*
- (b) *The local ring $\mathcal{O}_{\mathcal{X},\xi}$ is a discrete valuation ring for every generic point $\xi \in \mathcal{X}_s$.*

Proof. (cf. [AW12, Lemma 2.1]) By Serre's criterion for normality [Liu02, Theorem 8.2.23], we need to check the following properties:

(R₁) $\mathcal{X}$ is regular at x when $\text{codim } x \leq 1$, and

(S₂) $\text{depth } \mathcal{O}_{\mathcal{X},x} \geq \min\{2, \dim \mathcal{O}_{\mathcal{X},x}\}$

for all $x \in \mathcal{X}$.

The points on $\mathcal{X}$ with codimension 1 are precisely the closed points of the generic fibre, which is regular by assumption, and the generic points of the irreducible components of the special fibre. The codimension-one local rings under consideration are one-dimensional Noetherian local domains. Such a ring is regular if and only if it is a discrete valuation ring, so (R₁) is equivalent to (b). Condition (S₂) holds automatically at points of the generic fibre and at the generic points of $\mathcal{X}_s$. At every non-generic point $x \in \mathcal{X}_s$, it is equivalent to $\text{depth } \mathcal{O}_{\mathcal{X},x} \geq 1$ by [Liu02, Proposition 8.2.11], which in turn is equivalent to x not being an embedded associated point. $\square$

We discuss the important example of *plane models*. Let $X \subset \mathbb{P}_K^2$ be a smooth projective plane curve. Let $\mathcal{X}$ be the Zariski closure of X inside $\mathbb{P}_{\mathcal{O}_K}^2$. This is a proper and integral $\mathcal{O}_K$ -scheme with generic fibre X . By Remark 1.4 it is therefore a pre-model of X .

In concrete terms, X is defined by a homogeneous irreducible polynomial $F \in K[X, Y, Z]$. After multiplying F by a suitable element of $K^\times$, we may assume that F is integral (i.e. all coefficients are in $\mathcal{O}_K$) and primitive (not all coefficients are divisible by the uniformizer π). Then $\mathcal{X} \subset \mathbb{P}_{\mathcal{O}_K}^2$ is the hypersurface cut out by F . It follows that the special fibre is the plane curve $\mathcal{X}_s \subset \mathbb{P}_k^2$ cut out by the reduction $\bar{F} \in k[X, Y, Z]$ of F . In particular, the generic points $\xi \in \mathcal{X}_s$ correspond to the irreducible factors of $\bar{F}$.

Proposition 1.16. (i) *$\mathcal{X}$ is normal if and only if it is regular at every generic point $\xi \in \mathcal{X}_s$.*

- (ii) Let $\xi \in \mathcal{X}_s$ be a generic point, corresponding to an irreducible factor $\bar{G}$ of $\bar{F}$. Let r be the multiplicity of $\bar{G}$ in $\bar{F}$, choose a homogeneous lift $G \in \mathcal{O}_K[X, Y, Z]$, and write

$$F = G^r H_1 + \pi^e H_2, \quad (1.1)$$

where $e \geq 1$ is chosen so that $\bar{H}_2 \neq 0$ and $\bar{G} \nmid \bar{H}_1$. Then $\mathcal{X}$ is regular at ξ if and only if either $r = 1$, or $e = 1$ and $\bar{G} \nmid \bar{H}_2$.

Proof. (cf. [Liu02, Example 8.2.26]) The ambient scheme $\mathbb{P}_{\mathcal{O}_K}^2$ is regular, and $\mathcal{X}$ is a hypersurface in it. Thus $\mathcal{X}$ is an effective Cartier divisor in a regular scheme, so it is locally a complete intersection and therefore Cohen–Macaulay. Serre’s criterion then shows that $\mathcal{X}$ is normal if and only if it is regular at all its codimension-one points. Since X is smooth over K , the only possible nonregular points of codimension one are the generic points of $\mathcal{X}_s$. This proves (i).

Now let $\xi \in \mathcal{X}_s$ be a generic point. We may assume that ξ lies on the affine chart $Z \neq 0$. We dehomogenize with $x = X/Z$ and $y = Y/Z$ and consider $F, G, \dots$ as elements of $\mathcal{O}_K[x, y]$. Then ξ , as a point on the ambient space $\mathbb{A}_{\mathcal{O}_K}^2$, corresponds to the prime ideal $\mathfrak{p} := (\pi, G) \triangleleft \mathcal{O}_K[x, y]$. The localization of $\mathcal{O}_K[x, y]$ at $\mathfrak{p}$ is the local ring $R := \mathcal{O}_{\mathbb{A}_{\mathcal{O}_K}^2, \xi}$ with maximal ideal $\mathfrak{m} = (\pi, G)R$. Therefore, $\mathcal{O}_{\mathcal{X}, \xi} = R/(F)$.

By [Liu02, Corollary 4.2.12], $\mathcal{X}$ is regular at ξ if and only if $F \notin \mathfrak{m}^2$ which in turn is the case if and only if either $r = 1$, or $e = 1$ and $\bar{H}_2 \notin \bar{G}(x, y)k[x, y]$. $\square$

Example 1.17. (a) Consider the curve X defined by $y^2 = x^3 + px$ over $K = \mathbb{Q}_p$ with $p \neq 2$. In the setting of Proposition 1.16,

$$\mathcal{X} = \text{Proj } \mathcal{O}_K[X, Y, Z]/(X^3 + pXZ^2 - Y^2Z)$$

is a proper and flat $\mathcal{O}_K$ -scheme with $\mathcal{X}_\eta = X$. The special fibre is given by $\mathcal{X}_s = \text{Proj } k[X, Y, Z]/(X^3 - Y^2Z)$. On the affine chart $Z \neq 0$, its generic point ξ corresponds to the ideal $\mathfrak{m} = (p, x^3 - y^2)$. Here $r = e = 1$, so Proposition 1.16 shows that $\mathcal{X}$ is normal and therefore a model of X . Its special fibre is the integral plane curve with equation $y^2 = x^3$, which has a unique singularity at $(0, 0)$.

- (b) Let $X : y^3 = 1 + x^3 + 3x$ be a curve over $K = \mathbb{Q}_3$. The scheme

$$\mathcal{X} = \text{Proj } \mathcal{O}_K[X, Y, Z]/(Y^3 - Z^3 - X^3 - 3XZ^2) \quad (1.2)$$

has a non-reduced special fibre: On the affine chart $Z \neq 0$ we have $\mathcal{X}_s : 0 = y^3 - x^3 - 1 = (y - x - 1)^3$ over $\mathbb{F}_3$. In the notation of Proposition 1.16,

$$\begin{aligned} F(x, y) &= G(x, y)^r H_1(x, y) + 3H_2(x, y) \\ &:= (y - x - 1)^3 + 3(-x^2y + xy^2 + x^2 - 2xy + y^2 - y), \end{aligned}$$

so $r = 3$, but $e = 1$ and $\bar{H}_2(x, y) \notin \bar{G}(x, y)\mathbb{F}_3[x, y]$. Proposition 1.16 therefore shows that $\mathcal{X}$ is normal and hence a model of X . Its special fibre is a nonreduced hypersurface with multiplicity 3; its reduced subscheme $\mathcal{X}_s^{\text{red}}$ is the line $y - x - 1 = 0$.

The following example is the discrete-valuation-ring version of [Har77, Example III.9.8.4].

Example 1.18. Let p be an odd prime, let $K = \mathbb{Q}_p$, and let $\mathcal{O}_K = \mathbb{Z}_p$. Consider the morphism

$$\iota : \mathbb{P}_K^1 \longrightarrow \mathbb{P}_K^3, \quad [S : T] \longmapsto [T^2S - S^3 : T^3 - TS^2 : pTS^2 : S^3].$$

Since p is invertible in K , these four cubic forms give the cubic Veronese embedding after a linear change of coordinates. Thus, its image X is a smooth projective curve isomorphic to $\mathbb{P}_K^1$. Let $\mathcal{X}$ be the schematic closure of X in $\mathbb{P}_{\mathcal{O}_K}^3$. Then $\mathcal{X}$ is proper over $\mathcal{O}_K$ and integral. It is therefore flat by Remark 1.4, so $\mathcal{X}$ is a pre-model of X .

On the affine chart $W \neq 0$, set $x = X/W$, $y = Y/W$, and $z = Z/W$. With $t = T/S$, the generic fibre is parametrized by

$$x = t^2 - 1, \quad y = t^3 - t, \quad z = pt.$$

Eliminating t and taking the p -saturated closure gives

$$\mathcal{X} \cap \{W \neq 0\} = \operatorname{Spec} \mathbb{Z}_p[x, y, z]/I,$$

where

$$I = (p^2(x+1) - z^2, px(x+1) - yz, xz - py, y^2 - x^2(x+1)).$$

Hence the coordinate ring of the special fibre on this chart is

$$B = k[x, y, z]/(z^2, yz, xz, y^2 - x^2(x+1)).$$

Its reduced subscheme is the nodal cubic $z = 0, y^2 = x^2(x+1)$. Let $\mathfrak{m} = (x, y, z)$ be the maximal ideal of its node and let $\bar{z}$ denote the class of z in $B_{\mathfrak{m}}$. The defining relations give $\mathfrak{m}\bar{z} = 0$. Moreover, $\bar{z} \neq 0$: the homomorphism

$$B \longrightarrow k[\epsilon]/(\epsilon^2), \quad x, y \longmapsto 0, \quad z \longmapsto \epsilon,$$

extends to $B_{\mathfrak{m}}$ and has nonzero image on $\bar{z}$. It follows that $\operatorname{Ann}_{B_{\mathfrak{m}}}(\bar{z}) = \mathfrak{m}B_{\mathfrak{m}}$. Thus, the node is an embedded point of $\mathcal{X}_s$, and $\mathcal{X}$ is a pre-model that is not a model by Lemma 1.15.

1.5 Components and geometric valuations

We now associate a valuation of the function field to each irreducible component of the special fibre. This construction also relates the multiplicity of a component to the value group of its valuation.

Definition 1.19. Let F_X be the function field of X over K . A valuation $v : F_X \rightarrow \hat{\mathbb{R}}$ is a *type-II valuation* if

- (a) $v|_K = v_K$,
- (b) v is discrete, and
- (c) the residue field $\kappa(v)$ has transcendence degree 1 over k .

A valuation of this type is also known as a *geometric valuation*. We denote the set of all such valuations relative to K by

$$V_{\text{II}}(F_X/K).$$

When the base field is clear, we abbreviate this to $V_{\text{II}}(F_X)$. The number

$$m_K(v) := [\Gamma_v : \Gamma_K] \quad (1.3)$$

is called the *multiplicity of v relative to K* , where $\Gamma_v := v(F_X^*)$ is the value group. We write $m(v)$ when the base field is clear. We say that v is *weakly unramified over K* if $m_K(v) = 1$.

Remark 1.20. The multiplicity in Definition 1.19 is finite. Write $\Gamma_K = \gamma_K \mathbb{Z}$ with $\gamma_K > 0$. Since discreteness is part of the definition of a type-II valuation and $\Gamma_K \subseteq \Gamma_v$, there is a unique integer $m \geq 1$ such that

$$\Gamma_v = \frac{\gamma_K}{m} \mathbb{Z}.$$

Hence

$$[\Gamma_v : \Gamma_K] = m < \infty.$$

Example 1.21. The *Gauß valuation* on the polynomial ring $K[x]$ is defined as

$$v_G : K[x] \rightarrow \hat{\mathbb{R}}, \quad v_G \left(\sum_i a_i x^i \right) = \min_i (v_K(a_i)). \quad (1.4)$$

Since $\kappa(v_G) = k(\bar{x})$ (where $\bar{x}$ is the image of x in $\kappa(v_G)$), the Gauß valuation extends naturally and uniquely to $K(x)$, so $v_G \in V_{\text{II}}(K(x))$.

Let $\mathcal{X}$ be a model of X . We wish to assign a valuation in $V_{\text{II}}(F_X)$ to each generic point of the special fibre $\mathcal{X}_s$. Suppose that Z is an irreducible component of $\mathcal{X}_s$ with generic point ξ . Since ξ has codimension 1 in $\mathcal{X}$, the stalk $\mathcal{O}_{\mathcal{X},\xi}$ is a discrete valuation ring. Thus, we can define a valuation on F_X ,

$$v_Z : \text{Frac}(\mathcal{O}_{\mathcal{X},\xi}) = F_X \rightarrow \hat{\mathbb{R}}. \quad (1.5)$$

Since $\mathcal{X}$ is a flat $\mathcal{O}_K$ -scheme, $\mathcal{O}_{\mathcal{X},\xi}$ dominates $\mathcal{O}_K$, and we choose the unique scaling of v_Z whose restriction to K is v_K . We set

$$V(\mathcal{X}) := \{v_Z : Z \subset \mathcal{X}_s \text{ is an irreducible component}\}. \quad (1.6)$$

Proposition 1.22. *The valuation defined in (1.5) is a type-II valuation.*

Proof. As mentioned above, we can assume $v_Z|_K = v_K$. The residue field of the valuation is

$$\kappa(v_Z) = \kappa(\xi) = k(Z). \quad (1.7)$$

Since Z is an integral curve over k , it follows that $\text{trdeg}_k \kappa(v_Z) = 1$, showing $v_Z \in V_{\text{II}}(F_X)$. $\square$

Definition 1.23. The *multiplicity* of the irreducible component Z of $\mathcal{X}_s$ containing the generic point ξ is defined as

$$m(Z) := \text{length}_{\mathcal{O}_{\mathcal{X},\xi}}(\mathcal{O}_{\mathcal{X},\xi}) = \text{length}_{\mathcal{O}_{\mathcal{X},\xi}}(\mathcal{O}_{\mathcal{X},\xi}/\pi\mathcal{O}_{\mathcal{X},\xi}), \quad (1.8)$$

where π is a uniformising parameter of $\mathcal{O}_K$.

Proposition 1.24. *We have $m(v_Z) = m(Z)$.*

Proof. Since the model $\mathcal{X}$ is normal, the stalk $\mathcal{O}_{\mathcal{X},\xi}$ is a discrete valuation ring. Let π_ξ be a uniformising parameter of this ring. There is a unit $u \in \mathcal{O}_{\mathcal{X},\xi}^\times$ such that

$$\pi = u\pi_\xi^{m(Z)}. \quad (1.9)$$

Indeed, the exponent on the right is the length of $\mathcal{O}_{\mathcal{X},\xi}/\pi\mathcal{O}_{\mathcal{X},\xi} = \mathcal{O}_{\mathcal{X}_s,\xi}$. Write $\Gamma_K = \gamma_K\mathbb{Z}$ and choose π so that $v_K(\pi) = \gamma_K$. The displayed relation gives $v_Z(\pi_\xi) = \gamma_K/m(Z)$, and hence $\Gamma_{v_Z} = (\gamma_K/m(Z))\mathbb{Z}$. Hence

$$m(Z) = [\Gamma_{v_Z} : \Gamma_K] = m(v_Z). \quad (1.10)$$

$\square$

We finish the section by recording the standard facts about valuations in finite field extensions.

Proposition 1.25. *Let E/F be a finite extension of fields and let v be a discrete real-valued valuation on F . Then v has at least one and only finitely many extensions to E . For every extension w of v , set*

$$e(w/v) := [\Gamma_w : \Gamma_v] \quad \text{and} \quad f(w/v) := [\kappa(w) : \kappa(v)].$$

Then:

- (i) *The ramification index $e(w/v)$ and residue degree $f(w/v)$ are finite, and*

$$e(w/v)f(w/v) \leq [E : F].$$

- (ii) *Ramification indices and residue degrees are multiplicative in towers: if M/E is finite and u extends w to M , then*

$$e(u/v) = e(u/w)e(w/v) \quad \text{and} \quad f(u/v) = f(u/w)f(w/v).$$

- (iii) *The valuation w is discrete, and $\kappa(w)/\kappa(v)$ is finite. In particular, extension and restriction through a finite field extension preserve the transcendence degree of the residue field over any common subfield.*

Proof. Existence follows by applying [Aut26, Lemma 10.50.2] to the local subring $\mathcal{O}_v \subseteq E$; the intersection of the resulting valuation ring with F dominates $\mathcal{O}_v$ and hence equals it. For finiteness in the general case, let E_s be the maximal separable subextension of E/F . The extensions to E_s are the localizations at the finitely many maximal ideals of the integral closure of $\mathcal{O}_v$ in E_s ; see [Aut26, Remark 15.113.6]. Each of them extends uniquely through the purely inseparable extension E/E_s , because the relevant integral closure is a valuation ring by [Aut26, Lemma 15.123.3]. Thus there are only finitely many extensions to E .

For a fixed w , part (i) is [Aut26, Lemma 15.125.2], applied to $\mathcal{O}_v \subseteq \mathcal{O}_w$. Part (ii) follows from the multiplicativity of group indices and field degrees; compare [Aut26, Lemma 15.113.3]. If $\Gamma_v = \delta\mathbb{Z}$ and $e = e(w/v)$, then $e\Gamma_w \subseteq \Gamma_v$, so

$$\Gamma_w \subseteq \frac{\delta}{e}\mathbb{Z}.$$

This proves discreteness, while the remaining assertions in part (iii) follow from the finiteness of $f(w/v)$. $\square$

Corollary 1.26. *Let $F \subseteq E$ be a finite extension of one-variable function fields over K , let w be a real-valued valuation on E , and set $v = w|_F$.*

- (i) *Every extension to E of a valuation in $V_{\Pi}(F/K)$ belongs to $V_{\Pi}(E/K)$.*
- (ii) *If $w \in V_{\Pi}(E/K)$, then $v \in V_{\Pi}(F/K)$.*
- (iii) *Under either hypothesis,*

$$m_K(w) = e(w/v)m_K(v).$$

Proof. For part (i), Proposition 1.25 shows that w is discrete and that $\kappa(w)/\kappa(v)$ is finite. Thus $\kappa(w)$ and $\kappa(v)$ have the same transcendence degree over k , so w is type II. For part (ii), Γ_v is a nonzero subgroup of the discrete group Γ_w and is therefore discrete. Moreover, $\kappa(w)/\kappa(v)$ is finite by Proposition 1.25, so the two residue fields have the same transcendence degree over k . The formula in part (iii) is the identity

$$[\Gamma_w : \Gamma_K] = [\Gamma_w : \Gamma_v][\Gamma_v : \Gamma_K].$$

$\square$

If E contains a finite extension L/K and $w|_L = v_L$, then multiplicities relative to the two constant fields satisfy

$$m_K(w) = e(L/K)m_L(w), \quad e(L/K) := [\Gamma_L : \Gamma_K].$$

The finite set $V(\mathcal{X})$ motivates the classification of models developed in the next chapter.

Chapter 2

The correspondence between models and valuations

Building on the valuations associated with components in Chapter 1, this chapter proves that models of a fixed curve correspond to finite nonempty sets of geometric valuations of its function field. By a cover of nice curves we mean a finite dominant morphism. The proof is based on Chapter 3 of Julian R  th’s thesis [R  14]. The final section applies the correspondence to normalization in a finite cover, a viewpoint emphasized in R  th’s Proposition 5.2. For a Galois cover, the normalization carries the natural Galois action and the original model is recovered as its quotient.

2.1 Statement and interpretation of the correspondence theorem

The central result of this chapter says that the assignment $\mathcal{X} \mapsto V(\mathcal{X})$ classifies models by finite nonempty sets of geometric valuations and respects the natural partial orders.

Theorem 2.1. *The map*

$$\mathcal{X} \longmapsto V(\mathcal{X}) \tag{2.1}$$

defines an isomorphism of partially ordered sets between

- *the set of isomorphism classes of models of X , ordered by domination, where $\mathcal{X} \leq \mathcal{X}'$ if there is a modification $\mathcal{X}' \rightarrow \mathcal{X}$, and*
- *the set of finite, nonempty subsets of $V_{\text{II}}(F_X)$, ordered by inclusion.*

More precisely, for any two models $\mathcal{X}$ and $\mathcal{X}'$ of X , one has

$$\mathcal{X} \leq \mathcal{X}' \iff V(\mathcal{X}) \subseteq V(\mathcal{X}').$$

The proof occupies §§2.2–2.5.

2.2 Modifications and injectivity

We first prove the injective part of the correspondence theorem. Recall from Section 1.2 that models are ordered by $\mathcal{X} \leq \mathcal{X}'$ when there is a modification $\mathcal{X}' \rightarrow \mathcal{X}$, and that every modification is proper, dominant, and surjective by Proposition 1.8. We will show that these comparison maps induce inclusions between the corresponding sets of component valuations.

Lemma 2.2. *Let $\mathcal{X}$ be a model of X and let $\mathcal{X}'$ be a model of another irreducible smooth projective curve X' . Let $\varphi : \mathcal{X}' \rightarrow \mathcal{X}$ be a morphism of $\mathcal{O}_K$ -schemes which induces a cover $X' \rightarrow X$ on the generic fibre. Let ξ be a generic point of $\mathcal{X}_s$. Then the preimage of ξ under φ is nonempty and consists of generic points of $\mathcal{X}'_s$. If φ is birational, then the preimage consists of precisely one generic point.*

Proof. We follow the argument of [Rü14, Lemma 3.6]. The graph argument in the proof of Proposition 1.8 shows that φ is proper. Since its generic-fibre map is dominant, φ is dominant and therefore surjective. Let ξ' be any point above ξ . Dominance gives an injection $\kappa(\xi) \hookrightarrow \kappa(\xi')$, and $\text{trdeg}_k \kappa(\xi) = 1$. The point ξ' cannot be a closed point of the finite-type curve $\mathcal{X}'_s$, because the residue field of such a point is finite over k . It is therefore the generic point of an irreducible component of $\mathcal{X}'_s$. If φ is birational, then it is an isomorphism over the codimension-one point ξ by the standard fact used in Proposition 1.8. Hence there is exactly one point above ξ . $\square$

Rüth records the following consequence of the comparison of components and their valuations as [Rü14, Lemma 3.7].

Lemma 2.3. *Let $\varphi : \mathcal{X}' \rightarrow \mathcal{X}$ be a modification of models of X . Then $V(\mathcal{X}) \subseteq V(\mathcal{X}')$. In particular, the partial order $\mathcal{X} \leq \mathcal{X}'$ of models on X induces a partial order on the corresponding finite subsets of $V_{\text{II}}(F_X)$ via inclusion maps.*

Proof. This is a consequence of Proposition 1.22 and Lemma 2.2. $\square$

The converse comparison is stated in [Rü14, Lemma 3.8].

Proposition 2.4. *Let $\mathcal{X}$ and $\mathcal{X}'$ be models of X with $V(\mathcal{X}) \subseteq V(\mathcal{X}')$. Then there is a modification $\mathcal{X}' \rightarrow \mathcal{X}$.*

Proof. If $V(\mathcal{X}) \subseteq V(\mathcal{X}')$, then for every valuation $v_Z \in V(\mathcal{X})$ corresponding to an irreducible component $Z \subseteq \mathcal{X}_s$ with generic point ξ , there is an irreducible component $Z' \subseteq \mathcal{X}'_s$ with generic point ξ' , such that $\mathcal{O}_{\mathcal{X},\xi} = \mathcal{O}_{\mathcal{X}',\xi'} \subseteq F_X$. Together with the identification $\mathcal{X}_\eta = X = \mathcal{X}'_\eta$, this induces a birational map $\mathcal{X}' \dashrightarrow \mathcal{X}$ (cf. for example [Aut26, Lemma 29.50.2.]). By [Liu02, Theorem 8.3.20], this map is a morphism. Since it identifies $\mathcal{X}_\eta$ and $\mathcal{X}'_\eta$, it is a modification. $\square$

Rüth states the resulting injectivity assertion as [Rü14, Proposition 3.9] and cites [Liu02, Corollary 8.3.23] as its underlying source.

Corollary 2.5. *Two models $\mathcal{X}$ and $\mathcal{X}'$ of X are isomorphic as models of X if and only if $V(\mathcal{X}) = V(\mathcal{X}')$. In particular, the map defined in (2.1) is injective.*

Proof. Suppose that $V(\mathcal{X}) = V(\mathcal{X}')$. Then there exist morphisms $\mathcal{X} \rightarrow \mathcal{X}'$ and $\mathcal{X}' \rightarrow \mathcal{X}$. Each composition agrees with the identity on the dense generic fibre. Since the models are integral and separated, the compositions are therefore the identity morphisms. Thus the two morphisms are mutually inverse. $\square$

Example 2.6. We continue the discussion of plane models from Proposition 1.16. Let $F \in \mathbb{Z}_p[X, Y, Z]$ be homogeneous, irreducible, and primitive, and assume that the resulting plane pre-model

$$\mathcal{X} = \text{Proj } \mathbb{Z}_p[X, Y, Z]/(F) \quad (2.2)$$

of the plane projective curve X over $\mathbb{Q}_p$ is normal, for instance by the criterion of Proposition 1.16. Thus $\mathcal{X}$ is a model. Suppose that $\bar{F} \in \mathbb{F}_p[X, Y, Z]$ is irreducible. Then $\mathcal{X}_s$ has precisely one irreducible component Z with generic point ξ corresponding to the ideal $(p, \bar{F})$. In particular, $V(\mathcal{X}) = \{v_Z\}$.

2.3 Centres and extraction of valuations

For the proof of Theorem 2.1, it remains to show the surjectivity of the map ((2.1)), i.e., for a given finite, non-empty set of valuations $V \subseteq V_{\text{II}}(F_X)$, there is a model $\mathcal{X}$ of X such that $V = V(\mathcal{X})$. We will prove this in two steps, which we state in the following proposition. The extraction statement is based on [Rü14, Proposition 3.14], while the contraction statement is [Rü14, Proposition 3.16], whose proof invokes [Liu02, Theorem 8.3.36].

Proposition 2.7. (i) *Given a model $\mathcal{X}$ and a valuation $v \in V_{\text{II}}(F_X)$, there exists a modification $\varphi : \mathcal{X}' \rightarrow \mathcal{X}$ such that $v \in V(\mathcal{X}')$.*
 (ii) *Given a model $\mathcal{X}$ and a nonempty subset $V \subseteq V(\mathcal{X})$, there exists a modification $\varphi : \mathcal{X} \rightarrow \mathcal{X}'$ such that $V = V(\mathcal{X}')$.*

After proving Proposition 2.7, the proof is complete: Starting with a model $\mathcal{X}$ of X , we can apply part (i) for every $v \in V \subseteq V_{\text{II}}(F_X)$ to obtain a new model that induces all valuations in V , but possibly more. We then apply part (ii) to construct a model that induces exactly the valuations in V .

We fix a model $\mathcal{X}$ of X and a type-II valuation $v \in V_{\text{II}}(F_X)$.

Definition 2.8. A point $x \in \mathcal{X}$ is called a *centre* of v if there exists a morphism $f : \text{Spec } \mathcal{O}_v \rightarrow \mathcal{X}$ of $\mathcal{O}_K$ -schemes which maps the closed point to x and makes the following diagram commute:

$$\begin{array}{ccc} \text{Spec } \mathcal{O}_v & \xrightarrow{f} & \mathcal{X} \\ & \nwarrow \quad \nearrow & \\ & \text{Spec } F_X & \end{array}$$

Here, the left map is induced by the inclusion $\mathcal{O}_v \hookrightarrow \text{Frac } \mathcal{O}_v = F_X$, and the right map comes from the identification of X with the generic fibre $\mathcal{X}_\eta$. Specifically, it is the canonical map $\text{Spec } F_X \cong \text{Spec } \mathcal{O}_{\mathcal{X},\zeta} \rightarrow \mathcal{X}$, where ζ is the generic point of X .

Lemma 2.9. *There exists a unique centre $x \in \mathcal{X}$ of v .*

Proof. By the valuative criterion for properness, such a centre exists, and the morphism f is unique (see also [Aut26, Section 01J5, Lemma 01J6]):

$$\begin{array}{ccc} \text{Spec } F_X & \xrightarrow{\quad} & \mathcal{X} \\ \downarrow & \searrow f & \downarrow \\ \text{Spec } \mathcal{O}_v & \xrightarrow{\quad} & \text{Spec } \mathcal{O}_K. \end{array}$$

□

The following criterion is the observation made in [Rü14, Example 3.11].

Lemma 2.10. *We have $v \in V(\mathcal{X})$ if and only if the centre x of v is of codimension one. In that case x is the generic point on the special fibre which induces v .*

Proof. Assume $v \in V(\mathcal{X})$. Then v is induced by a generic point $\xi \in \mathcal{X}_s$ on the special fibre. The canonical morphism $\text{Spec } \mathcal{O}_{\mathcal{X},\xi} \rightarrow \mathcal{X}$ shows that v has centre ξ , which is indeed a point of codimension one. Conversely, suppose $v \in V_{\text{II}}(F_X)$ has centre $x \in \mathcal{X}$ of codimension one. The morphism $\text{Spec } \mathcal{O}_v \rightarrow \mathcal{X}$ uniquely factors through $\text{Spec } \mathcal{O}_v \rightarrow \text{Spec } \mathcal{O}_{\mathcal{X},x}$, so $\mathcal{O}_v$ dominates $\mathcal{O}_{\mathcal{X},x}$. Both are valuation rings in the common fraction field F_X , and $\mathcal{O}_v$ dominates $\mathcal{O}_{\mathcal{X},x}$. A valuation ring is maximal among local subrings of its fraction field under domination, so the domination cannot be proper; see [Aut26, Definition 10.50.1]. Hence $\mathcal{O}_v = \mathcal{O}_{\mathcal{X},x}$. Thus $v \in V(\mathcal{X})$. □

The next lemma combines [Rü14, Lemmas 3.12 and 3.13] in a form convenient for the extraction argument.

Lemma 2.11. *Let $x \in \mathcal{X}$ be the centre of v , and let $f \in \mathcal{O}_v^* \subset F_X$ be such that the reduction $\bar{f} \in \kappa(v)$ is transcendental over k . Then:*

- (i) *If $f \in \mathcal{O}_{\mathcal{X},x}$, then $v \in V(\mathcal{X})$.*
- (ii) *If the rational map $\mathcal{X} \dashrightarrow \mathbb{P}_{\mathcal{O}_K}^1$ induced by f is defined at the centre x of v , then $v \in V(\mathcal{X})$.*

Proof. (i) By Lemma 2.10, it suffices to show that x is a point of codimension one. Choose an open affine set $\text{Spec } A = U \subset \mathcal{X}$ such that $x \in U$ and $f \in A$. Denote by $\mathfrak{p} \subset A$ the prime ideal corresponding to x . Since v is non-trivial, the centre x is not the generic point of $\mathcal{X}$, and so the codimension is not zero. If the dimension of $A/\mathfrak{p}$ is at least one, then the inequality

$$\dim(A_{\mathfrak{p}}) + \dim(A/\mathfrak{p}) \leq \dim(A) = 2$$

implies

$$\text{codim}(\overline{\{x\}}, \mathcal{X}) = \dim(A_{\mathfrak{p}}) \leq 2 - \dim(A/\mathfrak{p}) \leq 2 - 1 = 1,$$

and hence equals one. Thus, it suffices to show that $A/\mathfrak{p}$ is not a field. If $A/\mathfrak{p}$ were a field, it would be a finite extension of k , but this is impossible since $\bar{f} \in A/\mathfrak{p} \subset \kappa(v)$ is transcendental over k .

- (ii) It suffices to show, using part (i), that $f \in \mathcal{O}_{\mathcal{X},x}$. Let $P = V(f^{-1})$ and $Z = V(f)$ be the sets of poles and zeros of f , respectively. Then the rational map $\mathcal{X} \dashrightarrow \mathbb{P}_{\mathcal{O}_K}^1$ is obtained by gluing the morphisms

$$\mathcal{X} \setminus P \rightarrow \text{Spec } \mathcal{O}_K[t] \subset \mathbb{P}_{\mathcal{O}_K}^1, \quad \text{and} \quad \mathcal{X} \setminus Z \rightarrow \text{Spec } \mathcal{O}_K[t^{-1}] \subset \mathbb{P}_{\mathcal{O}_K}^1,$$

which are induced by

$$\begin{array}{ccc} \mathcal{O}_K[t] \rightarrow \Gamma(\mathcal{X} \setminus P, \mathcal{O}_{\mathcal{X}}), & \text{and} & \mathcal{O}_K[t^{-1}] \rightarrow \Gamma(\mathcal{X} \setminus Z, \mathcal{O}_{\mathcal{X}}), \\ t \mapsto f & & t^{-1} \mapsto f^{-1} \end{array}$$

and is defined on $U := (\mathcal{X} \setminus P) \cup (\mathcal{X} \setminus Z) = \mathcal{X} \setminus (P \cap Z)$. By assumption, $x \in U$. If $x \in \mathcal{X} \setminus P$, the claim follows immediately, as

$$f \in \Gamma(\mathcal{X} \setminus P, \mathcal{O}_{\mathcal{X}}) \subset \mathcal{O}_{\mathcal{X},x}.$$

If $x \in \mathcal{X} \setminus Z$, then

$$f^{-1} \in \Gamma(\mathcal{X} \setminus Z, \mathcal{O}_{\mathcal{X}}) \subset \mathcal{O}_{\mathcal{X},x}.$$

The image of f^{-1} under the local ring homomorphism $\mathcal{O}_{\mathcal{X},x} \rightarrow \mathcal{O}_v$ is a unit by assumption, hence f^{-1} is also a unit in $\mathcal{O}_{\mathcal{X},x}$. Consequently $f \in \mathcal{O}_{\mathcal{X},x}$. $\square$

We can now finish the proof of Proposition 2.7(i).

Proof of Proposition 2.7(i). Let $f \in \mathcal{O}_v^\times \subset F_X$ be such that the reduction $\bar{f} \in \kappa(v)$ is transcendental over k . We also write $f : \mathcal{X} \dashrightarrow \mathbb{P}_{\mathcal{O}_K}^1$ for the rational map induced by f , and let U be its domain of definition. Define $\Gamma_f \subset \mathcal{X} \times_{\text{Spec } \mathcal{O}_K} \mathbb{P}_{\mathcal{O}_K}^1$ to be the graph of f . Recall that it is defined to be the closure of the graph $f|_U$ in $\mathcal{X} \times_{\text{Spec } \mathcal{O}_K} \mathbb{P}_{\mathcal{O}_K}^1 \supset U \times_{\text{Spec } \mathcal{O}_K} \mathbb{P}_{\mathcal{O}_K}^1$. This is almost a model of X , as it satisfies all the properties of a model except that it might not be normal.

$$\begin{array}{ccccc} & & \mathcal{X}' & & \\ & & \downarrow & & \\ \mathcal{X} \times_{\text{Spec } \mathcal{O}_K} \mathbb{P}_{\mathcal{O}_K}^1 & \supseteq & \Gamma_f & \longrightarrow & \mathbb{P}_{\mathcal{O}_K}^1 \\ & & \downarrow & \nearrow f & \\ & & \mathcal{X} & & \end{array}$$

The first projection $\mathcal{X} \times_{\text{Spec } \mathcal{O}_K} \mathbb{P}_{\mathcal{O}_K}^1 \rightarrow \mathcal{X}$ induces a birational morphism $\Gamma_f \rightarrow \mathcal{X}$. The second projection induces a morphism $\Gamma_f \rightarrow \mathbb{P}_{\mathcal{O}_K}^1$ which extends $f : \Gamma_f \dashrightarrow \mathbb{P}_{\mathcal{O}_K}^1$. Let $\mathcal{X}' \rightarrow \Gamma_f$ denote the normalization of Γ_f . This gives a model $\mathcal{X}'$ of X for which $f : \mathcal{X}' \rightarrow \mathbb{P}_{\mathcal{O}_K}^1$ is a morphism. By Lemma 2.11(ii), this implies that $v \in V(\mathcal{X}')$. $\square$

2.4 Contraction of components

The extraction construction may introduce component valuations outside the prescribed set. We now prepare the converse operation, which contracts precisely the components corresponding to these unwanted valuations.

We fix a model $\mathcal{X}$ of X and a nonempty subset $V \subseteq V(\mathcal{X})$. Denote by $\mathcal{E}$ the set of irreducible (integral) components of $\mathcal{X}_s$ that induce the valuations in $V(\mathcal{X}) \setminus V$.

Definition 2.12. A *contraction* of $\mathcal{E}$ is a model $\mathcal{X}'$ together with a modification $\varphi : \mathcal{X} \rightarrow \mathcal{X}'$ such that, for every integral vertical curve E on $\mathcal{X}_s$, the set $\varphi(E)$ is a point if and only if $E \in \mathcal{E}$.

It can be shown that if a contraction exists, it is unique up to unique isomorphism. Our goal now is to show that a contraction of $\mathcal{E}$ exists. Once this is done, the proof is complete, since for the contraction $\varphi : \mathcal{X} \rightarrow \mathcal{X}'$, we have by construction $V(\mathcal{X}') = V$.

For the two ingredients of the following criterion, see [Liu02, Lemmas 8.3.29 and 8.3.3].

Lemma 2.13. Let $\mathcal{L}$ be an invertible sheaf on $\mathcal{X}$ generated by global sections $s_0, \dots, s_n$. Consider the morphism $f : \mathcal{X} \rightarrow \mathbb{P}_{\mathcal{O}_K}^n$ associated to these sections. Let E be a vertical curve of $\mathcal{X}_s$ with the reduced induced closed subscheme structure. Then $f(E)$ is a single point if and only if $\mathcal{L}|_E \cong \mathcal{O}_E$.

For the contraction criterion in the form used here, see [Liu02, Proposition 8.3.30].

Proposition 2.14. The following are equivalent.

- (a) The contraction $\varphi : \mathcal{X} \rightarrow \mathcal{X}'$ of $\mathcal{E}$ exists.
- (b) There exists a Cartier divisor D on $\mathcal{X}$ such that $\deg(D|_{\mathcal{X}_\eta}) > 0$, that $\mathcal{L} \cong \mathcal{O}_{\mathcal{X}}(D)$ is generated by its global sections, and that for any integral vertical curve E , we have $\mathcal{L}|_E \cong \mathcal{O}_E$ if and only if $E \in \mathcal{E}$.

Proof. We only sketch (b) $\implies$ (a). The converse and the full proof are given in the cited proposition.

As $\mathcal{X}_\eta$ is an integral projective curve over a field, $\mathcal{L}|_{\mathcal{X}_\eta}$ is ample ([Liu02, Proposition 7.5.5]). Replacing D by a sufficiently large positive multiple (which does not

change the set E), we may choose finitely many global sections $s_0, \dots, s_n \in \Gamma(\mathcal{X}, \mathcal{L})$ which generate $\mathcal{L}$ on $\mathcal{X}$ and whose restrictions define a closed immersion of $\mathcal{X}_\eta$. Indeed, start with global generators of $\mathcal{L}$ and with sections of $\mathcal{L}|_{\mathcal{X}_\eta}$ defining a closed immersion; after multiplying the latter by a common power of a uniformiser, they extend across the special fibre, and the union of the two finite sets has both required properties.

Let $f : \mathcal{X} \rightarrow \mathbb{P}_{\mathcal{O}_K}^n$ be the morphism associated to these sections. Let Z be the closed subset $f(\mathcal{X}) \subset \mathbb{P}_{\mathcal{O}_K}^n$ endowed with the reduced (hence integral) scheme structure. Then f induces a dominant morphism $\mathcal{X} \rightarrow Z$ that we still denote by f . This is an isomorphism onto its image on the generic fibre by the choice of the sections. In particular, $f : \mathcal{X} \rightarrow Z$ is birational. Denote by $\pi : \mathcal{X}' \rightarrow Z$ the normalization morphism. As $\mathcal{X}$ is normal, f factors uniquely into $\varphi : \mathcal{X} \rightarrow \mathcal{X}'$ followed by $\pi : \mathcal{X}' \rightarrow Z$.

$$\begin{array}{ccc} & & \mathcal{X}' \\ & \nearrow \varphi & \downarrow \pi \\ \mathcal{X} & \xrightarrow{f} & Z \end{array}$$

Let E be an integral vertical curve on $\mathcal{X}$. Then

$$f(E) \text{ is a point} \iff \mathcal{L}|_E \simeq \mathcal{O}_E \iff E \in \mathcal{E},$$

where the first equivalence follows from Lemma 2.13 and the second holds by assumption. We claim that $\pi : \mathcal{X}' \rightarrow Z$ is a finite morphism. This completes the proof because $f(E)$ is a single point if and only if $\varphi(E)$ is a point. The morphism f is projective ([Liu02, Corollary 3.3.32(e)]), so $\mathcal{A} \cong f_* \mathcal{O}_{\mathcal{X}}$ is a coherent sheaf on Z ([Liu02, Corollary 5.3.5]). Furthermore, $\mathcal{A}$ is a sheaf of integrally closed algebras because $\mathcal{X}$ is normal. As a result, the normalization morphism $\pi : \mathcal{X}' \rightarrow Z$ coincides with the canonical morphism $\text{Spec } \mathcal{A} \rightarrow Z$ and is a finite morphism ([Liu02, Exercise 5.1.17]). $\square$

For the construction of a suitable Cartier divisor, we use the global-generation statement [Liu02, Lemma 8.3.32].

Lemma 2.15. *Let D be an effective horizontal Cartier divisor on $\mathcal{X}$, and $\mathcal{L} = \mathcal{O}_{\mathcal{X}}(D)$. Then there exists $m_0 \geq 1$ such that $\mathcal{L}^{\otimes m}$ is generated by its global sections for every $m \geq m_0$.*

Proposition 2.14 and Lemma 2.15 show that for a contraction morphism to exist, it suffices to find an effective Cartier divisor that meets the closed fibres in a suitable way. To construct such a divisor, it is important that the ring $\mathcal{O}_K$ is complete, which implies it is Henselian. For the equivalent formulations used below, see [Liu02, 8.3.33] and [Aut26, Lemma 04GG].

Definition 2.16. Let A be a Noetherian local ring with maximal ideal $\mathfrak{m}$ and residue field $k := A/\mathfrak{m}$. We say that A is Henselian if one of the following equivalent conditions holds:

- (a) Hensel's Lemma holds: For every monic $f \in A[T]$, and every simple root $a_0 \in k$ of $\bar{f} \in k[T]$, there exists a lift $a \in A$ (so $\bar{a} = a_0$) such that $f(a) = 0$.
- (b) Every finite A -algebra is a direct sum of local A -algebras.

For the following existence statement, see [Liu02, Lemma 8.3.35].

Lemma 2.17. *Let $x \in \mathcal{X}_s$ be a closed point on the special fibre. Then the following assertions hold.*

- (i) *There exists an irreducible horizontal Weil divisor Δ containing x .*
- (ii) *There exists an effective horizontal Cartier divisor D such that $\mathcal{X}_s \cap \text{Supp } D = \{x\}$.*

Proof. (i) Let $\mathfrak{m}_x$ be the maximal ideal of $\mathcal{O}_{\mathcal{X},x}$ and denote by π a uniformising parameter of $\mathcal{O}_K$. Let $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ be the prime ideals of $\mathcal{O}_{\mathcal{X},x}$ that are minimal among those containing $\pi\mathcal{O}_{\mathcal{X},x}$. As $\dim \mathcal{O}_{\mathcal{X},x} = 2$ ([Liu02, Proposition 8.3.4]), Krull's principal ideal theorem implies that $\mathfrak{m}_x \neq \mathfrak{p}_i$. Hence there exists

$$f \in \mathfrak{m}_x \setminus (\cup_i \mathfrak{p}_i).$$

Thus f is a regular element and $V(f)$ does not contain any irreducible component of $\text{Spec } \mathcal{O}_{\mathcal{X},x}$.

There exists an affine open subset $W \ni x$ and a regular element $g \in \mathcal{O}_{\mathcal{X}}(W)$ such that $g_x = f$ and $V(g) \cap W_s = \{x\}$. Let Δ be the closure in $\mathcal{X}$ of an irreducible component of $V(g)$ passing through x . Then Δ is an irreducible horizontal divisor passing through x .

- (ii) Let $E := V(g)$ and let W be as in (i). Then E is an effective Cartier divisor on W . Moreover, $E \rightarrow \text{Spec } \mathcal{O}_K$ is quasi-projective and quasi-finite because $\dim E_s < \dim \mathcal{X}_s = 1$. By a corollary of Zariski's Main Theorem ([Liu02, Corollary 4.4.8]), there exists an affine open neighbourhood U of x such that $E \cap U$ is an open subscheme of a scheme Z that is finite over $\text{Spec } \mathcal{O}_K$.

$$E \cap U \xrightarrow{\text{open emb.}} Z \xrightarrow{\text{finite}} \text{Spec } \mathcal{O}_K$$

Let D be the connected component of Z containing x . Since $\mathcal{O}_K$ is a Henselian local ring, Z is a disjoint union of local schemes. Hence D is local, and consequently $D \subseteq E \cap U \subseteq \mathcal{X}$.

Finally, since D is closed in Z , it is finite over $\text{Spec } \mathcal{O}_K$. A finite morphism $D \rightarrow \text{Spec } \mathcal{O}_K$ between affine schemes is projective ([Liu02, Exercise 3.3.22]), and therefore proper ([Liu02, Theorem 3.3.30]). The morphism $D \rightarrow \text{Spec } \mathcal{O}_K$ and the structural morphism $\mathcal{X} \rightarrow \text{Spec } \mathcal{O}_K$ are proper. Since $\mathcal{X}$ is separated over $\text{Spec } \mathcal{O}_K$, it follows that $D \rightarrow \mathcal{X}$ is proper by [Liu02, Proposition 3.3.16(e)]. Hence its image is closed in $\mathcal{X}$.

The scheme D is an open-and-closed component of the effective Cartier divisor $E \cap U$ on U . After shrinking to an open neighbourhood $U' \subseteq U$ of D , we may arrange that

$$E \cap U' = D.$$

Thus D is an effective Cartier divisor on U' . Since D is closed in $\mathcal{X}$, the open sets U' and $\mathcal{X} \setminus D$ cover $\mathcal{X}$, and this Cartier divisor glues with the empty divisor on $\mathcal{X} \setminus D$. The result is an effective horizontal Cartier divisor on $\mathcal{X}$ whose support meets $\mathcal{X}_s$ only at x .

□

2.5 Completion of the proof and consequences

Now we can finally prove Proposition 2.7(ii).

Proof of Proposition 2.7(ii). Let $Z_1, \dots, Z_n$ be the irreducible components of $\mathcal{X}_s$ that induce the valuations in $V \subseteq V(\mathcal{X})$. By Lemma 2.17(ii), there exists an effective horizontal Cartier divisor D_i such that $\mathcal{X}_s \cap \text{Supp } D_i$ is a point of Z_i that belongs to none of the components of $\mathcal{E}$. Let $D = \sum_{i=1}^n D_i$. By Lemma 2.15, after replacing D by a multiple if necessary, we may assume that $\mathcal{L} := \mathcal{O}_{\mathcal{X}}(D)$ is generated by its global sections.

The theorem now follows from Proposition 2.14. Indeed, $\mathcal{L}|_E \simeq \mathcal{O}_E$ if $E \in \mathcal{E}$ since $E \cap \text{Supp } D = \emptyset$; if $E \notin \mathcal{E}$, then E is some Z_i and $\deg \mathcal{L}|_E \geq \deg \mathcal{O}_{\mathcal{X}}(D_i)|_{Z_i} > 0$, and hence $\mathcal{L}|_E \not\simeq \mathcal{O}_E$. □

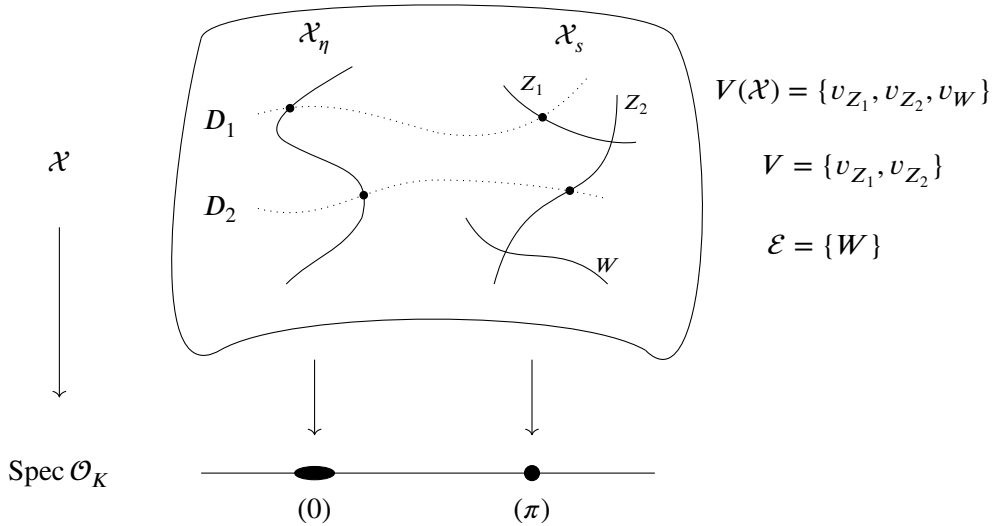


Figure 2.1: Illustration of the proof of Proposition 2.7(ii)

Remark 2.18. The proof of Proposition 2.7(ii) shows that all models $\mathcal{X}$ of X are in fact projective. Indeed, it suffices to choose $V = V(\mathcal{X})$ in the proof to obtain an effective Cartier divisor D that meets all the irreducible components of the fibre $\mathcal{X}_s$. Since $\mathcal{X}_s$ is a projective curve ([Liu02, Exercise 7.5.4]), $\mathcal{O}_{\mathcal{X}}(D)|_{\mathcal{X}_s}$ is ample ([Liu02, Proposition 7.5.5]). It follows from [Liu02, Corollary 5.3.24] that $\mathcal{O}_{\mathcal{X}}(D)$ is ample. Thus, $\mathcal{X} \rightarrow \operatorname{Spec} \mathcal{O}_K$ is projective ([Liu02, Proposition 5.3.6]).

The correspondence theorem therefore describes both the order structure on models and the construction of a model from its component valuations. We finish by explaining its functorial form for finite covers.

2.6 Normalization in finite covers

Let

$$\varphi : Y \longrightarrow X$$

be a cover of nice curves over K , and denote the induced finite extension of function fields by F_Y/F_X . If $\mathcal{X}$ is a model of X , let $\mathcal{Y}$ be the normalization of $\mathcal{X}$ in F_Y . For a valuation v on F_X , set

$$\operatorname{Ext}_{F_Y/F_X}(v) := \{w : w \text{ is a valuation on } F_Y \text{ and } w|_{F_X} = v\}.$$

By Corollary 1.26, every element of this set is a type-II valuation. The following proposition identifies these extensions geometrically and, in particular, proves that the set is finite and nonempty. It is Proposition 5.2 of R  th’s thesis [R  14], stated in the notation of this book.

Proposition 2.19 (Normalization and valuations). (i) *The scheme $\mathcal{Y}$ is a model of Y .*

(ii) *The cover φ extends uniquely to a finite morphism*

$$\mathcal{Y} \longrightarrow \mathcal{X}.$$

(iii) *Restriction of valuations induces a surjection*

$$V(\mathcal{Y}) \longrightarrow V(\mathcal{X}).$$

(iv) *For every $v \in V(\mathcal{X})$, the fibre over v is*

$$\operatorname{Ext}_{F_Y/F_X}(v).$$

Consequently,

$$V(\mathcal{Y}) = \bigsqcup_{v \in V(\mathcal{X})} \operatorname{Ext}_{F_Y/F_X}(v),$$

where the union is disjoint because a valuation on F_Y has a unique restriction to F_X . In particular, $\operatorname{Ext}_{F_Y/F_X}(v)$ is finite and nonempty for every $v \in V(\mathcal{X})$.

Proof. Since $\mathcal{X}$ is excellent, its normalization in the finite extension F_Y/F_X is finite over $\mathcal{X}$; see [Liu02, Theorem 8.2.39(d)]. Thus $\mathcal{Y}$ is proper over $\mathcal{O}_K$. It is normal and integral by construction, and it is flat over the discrete valuation ring $\mathcal{O}_K$ because it is integral. Its generic fibre is the normalization of X in F_Y , which is Y . This proves (i), and the normalization morphism gives the finite extension in (ii). Uniqueness follows because it is already determined on the dense generic fibre.

We explain the valuation statement, following the proof of [Rü14, Proposition 5.2]. Let $v \in V(\mathcal{X})$ and let ξ be the corresponding generic point of $\mathcal{X}_s$. The local ring $\mathcal{O}_{\mathcal{X},\xi}$ is the valuation ring $\mathcal{O}_v$. The generic points of $\mathcal{Y}_s$ above ξ correspond to the maximal ideals in the integral closure of $\mathcal{O}_v$ in F_Y . Their local rings are precisely the valuation rings of the extensions in $\text{Ext}_{F_Y/F_X}(v)$. Conversely, every generic point of $\mathcal{Y}_s$ lies over a generic point of $\mathcal{X}_s$ because the morphism $\mathcal{Y} \rightarrow \mathcal{X}$ is finite. Parts (iii) and (iv) now follow from the standard description of the integral closure of a discrete valuation ring in a finite extension. For a separable extension this is [Aut26, Remark 15.113.6]; the general case follows by passing through the maximal separable subextension and using [Aut26, Lemma 15.123.3] for the remaining purely inseparable extension, as in the proof of Proposition 1.25. $\square$

Definition 2.20. A cover $\varphi : Y \rightarrow X$ of nice curves is *Galois with group G* if the induced extension F_Y/F_X is Galois and

$$G = \text{Gal}(F_Y/F_X).$$

The action of G on F_Y corresponds to an action on Y over X .

Proposition 2.21 (Galois covers and quotients). *Let $\varphi : Y \rightarrow X$ be a Galois cover of nice curves over K , with group G . Let $\mathcal{X}$ be a model of X , and let $\mathcal{Y}$ be the normalization of $\mathcal{X}$ in F_Y . Then:*

- (i) *The action of G on Y , equivalently on F_Y , extends uniquely to an action of G on $\mathcal{Y}$ by $\mathcal{O}_K$ -automorphisms.*
- (ii) *The finite morphism $\mathcal{Y} \rightarrow \mathcal{X}$ is G -invariant.*
- (iii) *The quotient scheme exists, and the canonical morphism induces an isomorphism*

$$\mathcal{Y}/G \cong \mathcal{X}.$$

- (iv) *For every $v \in V(\mathcal{X})$, the group G acts transitively on*

$$\text{Ext}_{F_Y/F_X}(v).$$

Equivalently, it acts transitively on the irreducible components of $\mathcal{Y}_s$ lying above any fixed irreducible component of $\mathcal{X}_s$.

Proof. Let $U = \operatorname{Spec} A \subseteq \mathcal{X}$ be affine. Its inverse image in $\mathcal{Y}$ is affine, say $\operatorname{Spec} B$, where B is the integral closure of A in F_Y . Every $g \in G$ preserves B , because g fixes $A \subseteq F_X$ and therefore takes elements integral over A to elements integral over A . These actions agree on overlaps and extend the action on the generic fibre. They consequently define an action on $\mathcal{Y}$, and the extension is unique because the generic fibre is dense. The morphism $\mathcal{Y} \rightarrow \mathcal{X}$ is G -invariant because G fixes A on every such affine open.

We have $A \subseteq B^G$. Conversely, if $b \in B^G$, then

$$b \in F_Y^G = F_X = \operatorname{Frac}(A).$$

The element b is integral over the normal domain A , so $b \in A$. Thus $B^G = A$. The affine quotients $\operatorname{Spec} B^G$ glue, and the resulting identifications with the affine opens U identify $\mathcal{Y}/G$ with $\mathcal{X}$.

Finally, the elements of $\operatorname{Ext}_{F_Y/F_X}(v)$ correspond to the maximal ideals of the integral closure of the discrete valuation ring $\mathcal{O}_v$ in F_Y . The Galois group acts transitively on these maximal ideals by [Aut26, Lemma 15.114.1], proving part (iv). $\square$

Remark 2.22. The isomorphism $\mathcal{X} \cong \mathcal{Y}/G$ is a basic instance of descent for models: the model downstairs is recovered from the normalized model upstairs together with its Galois action. This does not assert that arbitrary G -equivariant models upstairs descend to models over $\mathcal{O}_K$. Moreover, the quotient statement concerns the total schemes; without additional hypotheses, one should not identify $\mathcal{X}_s$ with $\mathcal{Y}_s/G$, because formation of invariants need not commute with reduction in wild characteristic.

Example 2.23 (Realization as a cover of the projective line). Let Y be a nice curve over K . Since F_Y/K is a separably generated function field of transcendence degree one, it contains a separating transcendence element x such that $F_Y/K(x)$ is finite and separable. The element x defines a finite separable morphism

$$Y \longrightarrow X := \mathbb{P}_K^1.$$

Hence every model $\mathcal{X}$ of $X = \mathbb{P}_K^1$ gives a model $\mathcal{Y}$ of Y by normalization, and

$$V(\mathcal{Y}) = \bigsqcup_{v \in V(\mathcal{X})} \operatorname{Ext}_{F_Y/K(x)}(v).$$

For the standard model $\mathbb{P}_{\mathcal{O}_K}^1$, the special fibre has one component, whose valuation is the Gauß valuation on $K(x)$. This is the simplest instance of the description above. Chapter 5 develops Mac Lane valuations, which provide the starting point for explicit computations of such extensions.

Chapter 3

Base change and permanent models

This chapter studies models under finite extensions of the base field. The correspondence of Chapter 2 turns normalized base change into the extension of finitely many type-II valuations. For Galois extensions, the normalized base change carries a natural semilinear action and the original model is recovered as its quotient, while unramified base change is already normal and finite étale. Reducedness, permanence, and elimination of ramification can then be expressed in terms of their multiplicities.

Throughout this chapter, X is a smooth, projective and absolutely irreducible curve over the complete discretely valued field K satisfying Assumption 1.1.

3.1 Normalized base change and extensions of valuations

We first adapt Proposition 2.19 to an extension of the constant field. Let L/K be a finite extension. Since X is absolutely irreducible, X_L is integral and

$$F_{X_L} = F_X \otimes_K L$$

is a field. After choosing embeddings into a common overfield, this field may equivalently be viewed as the compositum $F_X \cdot L$.

Remark 3.1. Smoothness, projectivity, and absolute irreducibility are preserved under extension of the ground field, so X_L is a nice curve over L . When regarded as a K -scheme, however, X_L need not be absolutely irreducible over K . Thus the discussion below is the relative analogue of Proposition 2.19 rather than a literal special case of it.

The valuation v_K has a unique extension v_L to L because K is complete, and we always scale it so that

$$v_L|_K = v_K.$$

We do not renormalize v_L to have value group $\mathbb{Z}$. Thus, if $\Gamma_K = \mathbb{Z}$ and $e = e(L/K)$, then

$$\Gamma_L = \frac{1}{e}\mathbb{Z}.$$

Let $\mathcal{X}$ be a model of X . The ordinary base change is

$$\mathcal{X}_{\mathcal{O}_L} := \mathcal{X} \times_{\text{Spec } \mathcal{O}_K} \text{Spec } \mathcal{O}_L.$$

It is proper and flat over $\mathcal{O}_L$, with generic fibre X_L ; only normality may fail. We write

$$\mathcal{X}_L$$

for the normalized base change, namely the normalization of $\mathcal{X}_{\mathcal{O}_L}$ in F_{X_L} . It may also be obtained by normalizing $\mathcal{X}$ directly in the finite extension F_{X_L}/F_X . Indeed, on an affine open $\text{Spec } A \subseteq \mathcal{X}$, the integral closure of A in F_{X_L} contains $\mathcal{O}_L$, because every element of $\mathcal{O}_L$ is integral over $\mathcal{O}_K \subseteq A$. It therefore contains $A \otimes_{\mathcal{O}_K} \mathcal{O}_L$. Conversely, $A \otimes_{\mathcal{O}_K} \mathcal{O}_L$ is integral over A , so an element of F_{X_L} is integral over A if and only if it is integral over $A \otimes_{\mathcal{O}_K} \mathcal{O}_L$. Hence the two integral closures in F_{X_L} coincide. Finiteness of the normalization follows from excellence; see [Liu02, Theorem 8.2.39(d)].

For $v \in V_{\Pi}(F_X/K)$, set

$$\text{Ext}_L(v) := \{w \in V_{\Pi}(F_{X_L}/L) : w|_{F_X} = v\}. \quad (3.1)$$

Every extension of v to F_{X_L} restricts to v_L on L . Corollary 1.26 shows that it is discrete and that its residue field is finite over $\kappa(v)$; since the residue field l of L is finite over k , it has residue-field transcendence degree one over l and therefore belongs to the set in (3.1). For such an extension, the change-of-base formula from Chapter 1 reads

$$m_K(w) = e(L/K)m_L(w).$$

The following is the relative version of Proposition 2.19.

Proposition 3.2. *Restriction of valuations gives a surjection*

$$V(\mathcal{X}_L) \longrightarrow V(\mathcal{X}),$$

and the fibre over $v \in V(\mathcal{X})$ is $\text{Ext}_L(v)$. Thus

$$V(\mathcal{X}_L) = \bigsqcup_{v \in V(\mathcal{X})} \text{Ext}_L(v). \quad (3.2)$$

Proof. Let ξ be the generic point corresponding to $v \in V(\mathcal{X})$. The local ring $\mathcal{O}_{\mathcal{X},\xi}$ is the valuation ring $\mathcal{O}_v$. The generic points of the normalized base change above ξ correspond to the maximal ideals of the integral closure of $\mathcal{O}_v$ in F_{X_L} , and their local rings are the valuation rings of the elements of $\text{Ext}_L(v)$. This is the same discrete-valuation-ring argument as in Proposition 2.19. Every generic point of the special fibre maps to a generic point of $\mathcal{X}_s$ because the normalization morphism is finite. Hence every component valuation occurs in exactly one of the displayed fibres. $\square$

Computing normalized base change therefore amounts to understanding the extensions of each component valuation $v \in V(\mathcal{X})$ to F_{X_L} .

Suppose now that L/K is finite Galois, with group $G = \text{Gal}(L/K)$. Since X is geometrically integral, F_X and L are linearly disjoint over K . Consequently, $F_{X_L} = F_X \otimes_K L$ is a Galois extension of F_X with group G , acting by

$$g(f \otimes \lambda) = f \otimes g(\lambda).$$

Corollary 3.3 (Galois action on normalized base change). *Suppose that L/K is finite Galois with group G . Then:*

- (i) *The group G acts naturally on $\mathcal{X}_L$ by $\mathcal{O}_K$ -automorphisms, and this action is semilinear over $\mathcal{O}_L$. Thus, for every $g \in G$, the diagram*

$$\begin{array}{ccc} \mathcal{X}_L & \xrightarrow{g} & \mathcal{X}_L \\ \downarrow & & \downarrow \\ \text{Spec } \mathcal{O}_L & \xrightarrow{g} & \text{Spec } \mathcal{O}_L \end{array}$$

commutes.

- (ii) *The finite morphism $\mathcal{X}_L \rightarrow \mathcal{X}$ is G -invariant.*
- (iii) *The quotient scheme exists, and the canonical morphism induces an isomorphism*

$$\mathcal{X}_L/G \cong \mathcal{X}.$$

- (iv) *For every $v \in V(\mathcal{X})$, the group G acts transitively on $\text{Ext}_L(v)$. Equivalently, it acts transitively on the irreducible components of $(\mathcal{X}_L)_s$ lying above any fixed irreducible component of $\mathcal{X}_s$.*

Proof. The action already exists on the ordinary base change $\mathcal{X}_{\mathcal{O}_L}$: on an affine chart it is induced by

$$g(a \otimes \lambda) = a \otimes g(\lambda).$$

It preserves the integral closure of the coordinate ring in F_{X_L} and therefore extends to the normalization $\mathcal{X}_L$. The displayed formula also shows that the action is linear over $\mathcal{O}_K$, semilinear over $\mathcal{O}_L$, and leaves the morphism to $\mathcal{X}$ invariant.

The affine invariant-ring argument in Proposition 2.21 applies without requiring X_L to be a nice curve over K . Indeed, if $U = \text{Spec } A \subseteq \mathcal{X}$ and B is the integral closure of A in F_{X_L} , then

$$B^G = A$$

because $F_{X_L}^G = F_X$ and A is normal. These affine quotients glue and give $\mathcal{X}_L/G \cong \mathcal{X}$.

By Proposition 3.2, the set $\text{Ext}_L(v)$ corresponds to the generic points of $(\mathcal{X}_L)_s$ above the generic point associated with v . Equivalently, it corresponds to the maximal ideals in the integral closure of $\mathcal{O}_v$ in F_{X_L} . The transitivity assertion is therefore [Aut26, Lemma 15.114.1]. $\square$

As in Remark 2.22, the quotient is an assertion about the total schemes and does not, without further hypotheses, identify $\mathcal{X}_s$ with $(\mathcal{X}_L)_s/G$.

Proposition 3.4 (Unramified base change). *Let L/K be a finite unramified extension. Then the ordinary base change $\mathcal{X}_{\mathcal{O}_L}$ is normal, and hence*

$$\mathcal{X}_L = \mathcal{X}_{\mathcal{O}_L}.$$

Moreover, the canonical morphism

$$\mathcal{X}_L \longrightarrow \mathcal{X}$$

is finite étale.

Proof. Since L/K is a finite unramified extension of complete discretely valued fields, $\mathcal{O}_L$ is finite over $\mathcal{O}_K$ and $\mathcal{O}_K \rightarrow \mathcal{O}_L$ is unramified; see [Aut26, Remark 15.113.6 and Definition 15.113.7] and [Aut26, Lemma 41.3.3]. It is flat because $\mathcal{O}_L$ is torsion-free over the discrete valuation ring $\mathcal{O}_K$, and is therefore finite étale. Hence $\mathcal{X}_{\mathcal{O}_L} \rightarrow \mathcal{X}$ is finite étale by base change. Since $\mathcal{X}$ is normal, $\mathcal{X}_{\mathcal{O}_L}$ is normal by [Aut26, Proposition 41.19.7]. Thus ordinary base change already equals its normalization. $\square$

Remark 3.5. Finite unramified base change preserves and reflects regularity:

$$\mathcal{X} \text{ is regular} \iff \mathcal{X}_L \text{ is regular.}$$

Indeed, $\mathcal{X}_L \rightarrow \mathcal{X}$ is faithfully flat and finite étale, and regularity is preserved and reflected by étale local homomorphisms; see [Aut26, Proposition 41.19.4]. More generally, properties of locally Noetherian schemes that are local for the étale topology may be checked after finite unramified base change.

If l/k is the residue-field extension, then

$$(\mathcal{X}_L)_s = \mathcal{X}_s \times_{\text{Spec } k} \text{Spec } l.$$

Irreducible components may split after extending the residue field, but their multiplicities are unchanged. More precisely, for $v \in V(\mathcal{X})$ and $w \in \text{Ext}_L(v)$, the induced extension $\mathcal{O}_v \subseteq \mathcal{O}_w$ is étale and hence has ramification index $e(w/v) = 1$. Since also $e(L/K) = 1$, the multiplicity formulas from Chapter 1 give

$$m_L(w) = m_K(w) = e(w/v)m_K(v) = m_K(v).$$

Thus unramified extensions do not create a normalization problem and do not alter component multiplicities. The phenomenon studied below is therefore entirely associated with ramified extensions of the constant field.

Definition 3.6. Let $v \in V_{\Pi}(F_X/K)$. A finite extension L/K eliminates the ramification of v if

$$m_L(w) = 1$$

for every $w \in \text{Ext}_L(v)$. It eliminates the ramification of a finite set $V \subseteq V_{\Pi}(F_X/K)$ if it eliminates the ramification of every $v \in V$.

3.2 Reduced special fibres and permanence

The multiplicity criterion of Chapter 1 immediately detects reduced special fibres.

Proposition 3.7. *Let $\mathcal{X}$ be a model of X . The following conditions are equivalent:*

- (a) $\mathcal{X}_s$ is reduced;
- (b) $\mathcal{O}_{\mathcal{X}_s, \xi}$ is a field for every generic point $\xi \in \mathcal{X}_s$;
- (c) $m_K(v) = 1$ for every $v \in V(\mathcal{X})$.

Proof. The special fibre has no embedded points by Lemma 1.15, because $\mathcal{X}$ is normal. Hence it is reduced if and only if it is reduced at all its generic points, which is equivalent to condition (b). If Z is the component with generic point ξ , Proposition 1.24 gives

$$m_K(v_Z) = \text{length}_{\mathcal{O}_{\mathcal{X}_s, \xi}} \mathcal{O}_{\mathcal{X}_s, \xi}.$$

This length is one exactly when $\mathcal{O}_{\mathcal{X}_s, \xi}$ is a field. □

Definition 3.8. A model $\mathcal{X}$ of X is called *permanent* if $\mathcal{X}_{\mathcal{O}_L}$ is normal for every finite extension L/K .

Component valuations alone do not detect normality at closed points of an ordinary base change. The following proof therefore uses the normality criterion from Chapter 1.

Proposition 3.9. *If $\mathcal{X}_s$ is reduced, then $\mathcal{X}$ is permanent.*

Proof. Let L/K be finite, let l/k be the induced residue-field extension, and set $\mathcal{X}' = \mathcal{X}_{\mathcal{O}_L}$. Since k is perfect and $\mathcal{X}_s$ is reduced, the finite-type k -scheme $\mathcal{X}_s$ is geometrically reduced; see [Aut26, Lemma 33.6.3]. Thus

$$\mathcal{X}'_s = (\mathcal{X}_s)_l$$

is reduced and in particular has no embedded points.

Let ξ' be a generic point of $\mathcal{X}'_s$. The ring $A' = \mathcal{O}_{\mathcal{X}', \xi'}$ is a one-dimensional Noetherian local domain, and

$$A'/\pi_L A' = \mathcal{O}_{\mathcal{X}'_s, \xi'}$$

is a field. Hence the maximal ideal of A' is generated by the nonzero element π_L , so A' is a discrete valuation ring by [Aut26, Lemma 10.119.7]. Lemma 1.15 now shows that $\mathcal{X}'$ is normal. □

3.3 Epp's theorem and elimination of ramification

Epp's theorem supplies the extensions required in Definition 3.6. The original result is [Epp73]; a gap in one part of its proof was corrected in [Kuh03]. We use the solution form given by [Aut26, Proposition 15.118.8], whose hypotheses fit the local rings occurring here.

Theorem 3.10 (Epp: elimination of ramification). *Let $v \in V_{\Pi}(F_X/K)$. There exists a finite extension L/K that eliminates the ramification of v .*

Proof. By Theorem 2.1, the valuation v is a component valuation on some model of X . Its valuation ring $\mathcal{O}_v$ is therefore a discrete valuation ring essentially of finite type over $\mathcal{O}_K$. Apply [Aut26, Proposition 15.118.8] to the extension $\mathcal{O}_K \subset \mathcal{O}_v$. It gives a finite extension L/K such that every localization of the integral closure of $\mathcal{O}_v$ in F_{X_L} is weakly unramified over $\mathcal{O}_L$. These localizations are precisely the valuation rings of the elements of $\text{Ext}_L(v)$, so $m_L(w) = 1$ for every such w . $\square$

For completeness, the more general form of Epp's theorem requires, in positive residue characteristic p , that every element of $\bigcap_{n \geq 1} \kappa(v)^{p^n}$ be separable algebraic over k ; see [Aut26, Theorem 15.117.18]. The essentially-finite-type form used above removes the need to verify this condition separately.

Corollary 3.11. *For every finite subset $V \subseteq V_{\Pi}(F_X/K)$, there exists a finite extension L/K that eliminates the ramification of every $v \in V$.*

Proof. For each $v \in V$, choose a finite solution L_v/K as in the proof of Theorem 3.10, and let L be a finite extension containing all L_v . A solution remains a solution after every further finite extension by [Aut26, Lemma 15.118.1]. Thus L/K eliminates the ramification of every $v \in V$. $\square$

Corollary 3.12. *Given a model $\mathcal{X}$, there exists a finite extension L/K such that $\mathcal{X}_L$ has reduced special fibre and is permanent.*

Proof. Apply Corollary 3.11 to the finite set $V(\mathcal{X})$. Proposition 3.2 and Definition 3.6 show that every component valuation of $\mathcal{X}_L$ has multiplicity one relative to L . Its special fibre is therefore reduced by Proposition 3.7. The residue field of L is a finite extension of the perfect field k and is therefore perfect, so the standing hypotheses used in Proposition 3.9 remain valid over L . Applying that proposition over L shows that $\mathcal{X}_L$ is permanent. $\square$

3.4 Permanent models

The preceding results combine into the main theorem of the chapter.

Theorem 3.13. *Let $\mathcal{X}$ be a model of X .*

- (i) *The following conditions are equivalent:*

- (a) $\mathcal{X}$ is permanent;
 - (b) $\mathcal{X}_s$ is reduced;
 - (c) every $v \in V(\mathcal{X})$ is weakly unramified over K , equivalently $m_K(v) = 1$.
- (ii) There exists a finite extension L/K such that $\mathcal{X}_L$ is permanent.

Proof. Proposition 3.7 proves the equivalence of conditions (b) and (c). Proposition 3.9 proves that condition (b) implies condition (a). Part (ii) is Corollary 3.12.

Suppose now that $\mathcal{X}$ is permanent. Choose L/K as in Corollary 3.12 and set $\mathcal{Y} = \mathcal{X}_L$. Since $\mathcal{X}_{\mathcal{O}_L}$ is normal, its normalization map $\mathcal{Y} \rightarrow \mathcal{X}_{\mathcal{O}_L}$ is an isomorphism. Hence $(\mathcal{X}_s)_l = \mathcal{Y}_s$ is reduced, where l is the residue field of L . The extension l/k is faithfully flat, so reducedness descends to $\mathcal{X}_s$. This proves that condition (a) implies condition (b). $\square$

3.5 Abhyanka's lemma and tame elimination

Epp's theorem is needed in the presence of wild ramification. For tame component valuations, Abhyanka's lemma gives an explicit choice of base extension.

Definition 3.14. The extension of discrete valuation rings $\mathcal{O}_K \subseteq \mathcal{O}_v$ associated with $v \in V_{\text{II}}(F_X/K)$ is *tame* if $\kappa(v)/k$ is separable and either $\text{char } k = 0$, or $\text{char } k = p > 0$ and $p \nmid m_K(v)$. We call v *tame over K* when this extension of discrete valuation rings is tame.

The residue-field condition is automatic under the standing assumption that k is perfect, but it is included to make the hypothesis of Abhyanka's lemma explicit. The following statement is a direct application of [Aut26, Lemma 15.116.4].

Lemma 3.15 (Abhyanka). *Let $v \in V_{\text{II}}(F_X/K)$ be tame over K , and let L/K be a finite extension such that*

$$m_K(v) \mid e(L/K).$$

Then L/K eliminates the ramification of v .

Proof. Apply Abhyanka's lemma to the extension of discrete valuation rings $\mathcal{O}_K \subset \mathcal{O}_v$ and the finite extension L/K . Its hypotheses are exactly the tameness conditions in Definition 3.14 and the displayed divisibility. Every localization of the integral closure of $\mathcal{O}_v$ in F_{X_L} is then weakly unramified over $\mathcal{O}_L$. These localizations are the valuation rings of the elements $w \in \text{Ext}_L(v)$, and hence $m_L(w) = 1$ for all of them. $\square$

Corollary 3.16. *Let $V \subseteq V_{\text{II}}(F_X/K)$ be a finite set of tame valuations, and set*

$$e = \text{lcm}\{m_K(v) : v \in V\}.$$

If π_K is a uniformising parameter of K and $L = K(\pi_K^{1/e})$, then L/K is totally ramified of ramification index e and eliminates the ramification of every $v \in V$.

Proof. If $\text{char } k = p > 0$, then e is prime to p ; in residue characteristic zero there is no such restriction. The polynomial $T^e - \pi_K$ is Eisenstein, and its root generates a totally ramified extension of ramification index e ; see [Aut26, Lemma 15.116.2]. The result follows by applying Lemma 3.15 to every $v \in V$. $\square$

3.6 Wild examples

The following examples concern wild component valuations and illustrate that the ramification index of the constant-field extension alone does not determine whether ramification is eliminated.

Example 3.17 (Eliminating the ramification of a component). Let $K = \mathbb{Q}_3$ and consider the projective plane curve with affine equation

$$X : y^3 = 3x^2 + 1.$$

Its plane model $\mathcal{X}$ over $\mathbb{Z}_3$ is normal by Proposition 1.16. The special fibre is the triple line $y - 1 = 0$, so it has a unique component valuation v . Put $z = y - 1$. The equation

$$z^3 + 3z^2 + 3z - 3x^2 = 0$$

is Eisenstein over the Gauß valuation on $K(x)$. The component valuation v restricts to this Gauß valuation on $K(x)$. Hence $v(z) = 1/3$, $\Gamma_v = \frac{1}{3}\mathbb{Z}$, and

$$m_K(v) = 3.$$

Let $L = K(\theta)$ with $\theta^3 = 3$, and write $y = 1 + \theta y_1$. The equation becomes

$$y_1^3 + \theta^2 y_1^2 + \theta y_1 = x^2.$$

Its reduction is the integral curve $y_1^3 = x^2$, because x^2 is not a cube in $\mathbb{F}_3(x)$. The local ring at its generic point is a discrete valuation ring containing the corresponding local ring of the ordinary base change $\mathcal{X}_{\mathcal{O}_L}$ and integral over it, so it is the local normalization above the generic point of $\mathcal{X}_s$. In particular, there is one component above that point. Thus $\text{Ext}_L(v) = \{w\}$. Since the corresponding special-fibre component is reduced, Proposition 3.7 gives

$$m_L(w) = 1.$$

Therefore L/K eliminates the ramification of v , and $\mathcal{X}_L$ has reduced special fibre and is permanent.

Example 3.18 (Two wildly ramified quadratic extensions). Let $K = \mathbb{Q}_2$ and consider

$$X : y^2 = 8x^3 - 2.$$

Its projective equation is

$$Y^2 Z = 8X^3 - 2Z^3.$$

The plane model $\mathcal{X}$ is normal by Proposition 1.16. Its special fibre has the affine component Z given by $y = 0$ with multiplicity two and a component Z_∞ at infinity with multiplicity one. Let v be the component valuation associated with Z . Its restriction to $K(x)$ is the Gauß valuation. Since

$$y^2 = 2(4x^3 - 1),$$

where $4x^3 - 1$ is a unit at the generic point of Z , we have $v(y) = 1/2$ and

$$m_K(v) = 2.$$

This valuation is wild because the residue characteristic is two. The component valuation v_∞ associated with Z_∞ has $m_K(v_\infty) = 1$ and is tame. Lemma 3.15 shows that every finite extension considered below eliminates its ramification.

- (a) Let $L_1 = K(\theta_1)$ with $\theta_1^2 = 2$ and set

$$y = 2y_1 - \theta_1 - 2.$$

Then

$$y_1^2 - (\theta_1 + 2)y_1 = 2x^3 - (2 + \theta_1).$$

Relative to the valuation on $L_1(x)$, this quadratic is Eisenstein. It therefore defines the unique component valuation w_1 in the normalization above the generic point of Z , so $\text{Ext}_{L_1}(v) = \{w_1\}$, and its Newton polygon gives $w_1(y_1) = 1/4$. Since $\Gamma_{L_1} = \frac{1}{2}\mathbb{Z}$, it follows that

$$m_{L_1}(w_1) = 2.$$

Thus L_1/K does not eliminate the ramification of v , and the corresponding component occurs with multiplicity two in the normalized special fibre of $\mathcal{X}_{L_1}$. In particular, $\mathcal{X}_{L_1}$ is not permanent.

- (b) Let $L_2 = K(\theta_2)$ with $\theta_2^2 = -2$ and set

$$y = 2\theta_2 y_2 + \theta_2.$$

The equation becomes

$$y_2^2 + y_2 = -x^3.$$

Its reduction is the Artin–Schreier equation $y_2^2 + y_2 = x^3$. This polynomial is irreducible over $\mathbb{F}_2(x)$: an element of the form $a^2 + a$ cannot have a pole of odd order. It is also separable, so it defines the unique reduced component above Z . Thus $\text{Ext}_{L_2}(v) = \{w_2\}$. Hence

$$m_{L_2}(w_2) = 1.$$

The extension L_2/K eliminates the ramification of v . It also eliminates the ramification of v_∞ , as observed above. Proposition 3.2 therefore shows that all component valuations of $\mathcal{X}_{L_2}$ have multiplicity one relative to L_2 . Its special fibre is reduced by Proposition 3.7, and the model is permanent by Proposition 3.9.

The extensions L_1/K and L_2/K have the same ramification index, but only the second eliminates the ramification of v . This is precisely the behaviour not covered by the tame form of Abhyanka's lemma.

Part II

Analytic and computational tools

Stefan Wewers, Kletus Stern, and Maximilian Fülle

This part develops analytic and computational tools for describing and working with valuations arising from models. The material currently included, Chapter [5](#), is essentially the content of the last talk of the miniseminar from which this book originated. It begins the development with Mac Lane valuations, which provide explicit descriptions in terms of polynomials, augmentations, discs, and discoids. The remainder of Part II will be added later.

Chapter 5

Mac Lane valuations

In his seminal work [Mac36], S. Mac Lane considered and essentially solved the following problem. Let v_K be a nontrivial discrete valuation on a field K . What are the extensions of v_K to an (arbitrary) pseudovaluation v on the polynomial ring $K[x]$? How can we describe such a pseudovaluation explicitly? These results have been applied to the problem of constructing models of curves in J. R  th's thesis [R  14]. This chapter gives a brief overview of pseudovaluations and key polynomials, augmentations and inductive valuations, minimal augmentation chains, and their relation to discs and discoids. These constructions provide an explicit counterpart to the geometric valuations used in Part I.

Fix an algebraic closure $\overline{K}$ of K and an extension of v_K to $\overline{K}$, which we again denote by v_K .

5.1 Pseudovaluations and key polynomials

We first recall pseudovaluations on $K[x]$ and then single out the Mac Lane pseudovaluations and their key polynomials used in the subsequent constructions.

Definition 5.1. We call $v : K[x] \rightarrow \hat{\mathbb{R}}$ a pseudovaluation if $v(0) = \infty$, $v(1) = 0$, and

- (a) $v(fg) = v(f) + v(g)$,
- (b) $v(f + g) \geq \min\{v(f), v(g)\}$ for all $f, g \in K[x]$.

Remark 5.2. Let v be a pseudovaluation on $K[x]$.

- (a) The definition implies that $\mathfrak{p}_v := v^{-1}(\infty) < K[x]$ is a prime ideal and that v induces a valuation

$$\bar{v} : K[x]/\mathfrak{p}_v \rightarrow \hat{\mathbb{R}}. \quad (5.1)$$

If $\mathfrak{p}_v = (0)$, then $v = \bar{v}$ is a valuation in the usual sense.

- (b) Let $\varphi \in K[x]$ be monic and irreducible, and let $L = K[x]/(\varphi)$. Then we have a one-to-one correspondence between the valuations $v_L : L \rightarrow \hat{\mathbb{R}}$ extending v_K and the pseudovaluations $v : K[x] \rightarrow \hat{\mathbb{R}}$ extending v_K with $v(\varphi) = \infty$. It is given by the mutually inverse maps

$$v_L \mapsto \left(K[x] \rightarrow L \xrightarrow{v_L} \hat{\mathbb{R}} \right) \quad (5.2)$$

and

$$v \mapsto \left(L = K[x]/(\varphi) \xrightarrow{\bar{v}} \hat{\mathbb{R}} \right). \quad (5.3)$$

Definition 5.3. A pseudovaluation $v : K[x] \rightarrow \hat{\mathbb{Q}}$ is called a *Mac Lane pseudovaluation* if the following conditions hold.

- (c) $v|_K = v_K$,
- (d) $v(x) \geq 0$,
- (e) v is discrete, i.e. the subgroup generated by the finite values,

$$\Gamma_v := \langle v(f) : f \in K[x] \setminus \mathfrak{p}_v \rangle, \quad (5.4)$$

is equal to $\frac{1}{e_v} \mathbb{Z}$ for some positive integer e_v ,

- (f) if $\mathfrak{p}_v = (0)$, then the extension $k(v)|k$ has transcendence degree one.

We denote the set of all Mac Lane pseudovaluations of $K[x]$ by $V(K[x])^*$ and the subset of Mac Lane valuations of $K[x]$ by $V(K[x]) \subset V(K[x])^*$.

For the valuations in $V(K[x])$, there is a partial order: we say that $v \leq v'$ if and only if $v(f) \leq v'(f)$ for all $f \in K[x]$.

Example 5.4. The Gauß valuation $v_G \in V(K[x])^*$ is minimal with respect to this partial order.

Definition 5.5. Let $v \in V(K[x])$ be a Mac Lane valuation and let $f, g, h \in K[x]$.

- (a) We say that f and g are *v-equivalent* (written $f \sim_v g$) if $v(f - g) > v(f) = v(g)$.
- (b) The polynomial f is said to *v-divide* g (written $f \mid_v g$) if $g \sim_v fh$ for some $h \in K[x]$. It is called *v-irreducible* if $f \mid_v gh$ implies $f \mid_v g$ or $f \mid_v h$, and *v-minimal* if $f \mid_v g$ implies $\deg f \leq \deg g$.
- (c) A polynomial $\phi \in K[x]$ is called a *key polynomial* for v if ϕ is monic, integral, *v-irreducible* and *v-minimal*.

Remark 5.6. One immediate consequence of these definitions is that any key polynomial ϕ for v is irreducible (as an element of $K[x]$).

5.2 Augmentations and inductive valuations

Starting from a valuation and a key polynomial, augmentation produces a new pseudovaluation by prescribing a value for that polynomial which is at least its previous value. Iterating this construction leads to inductive valuations and their augmentation chains.

Let ϕ be a key polynomial for a valuation $v \in V(K[x])$ and let $\lambda \in \hat{\mathbb{Q}}$ with $\lambda \geq v(\phi)$. Then we can define a Mac Lane pseudovaluation

$$v' = [v, v'(\phi) = \lambda] \in V(K[x])^* \quad (5.5)$$

as follows. Any polynomial $f \in K[x]$ can be written uniquely as

$$f = \sum_i f_i \phi^i \quad (5.6)$$

with $\deg f_i < \deg \phi$. Then let

$$v'(f) := \min_i (v(f_i) + i \cdot \lambda). \quad (5.7)$$

By construction, $v \leq v'$, and $v < v'$ if and only if $v(\phi) < \lambda$.

Definition 5.7. The Mac Lane pseudovaluation v' defined in (5.5) is called the *augmentation* of v with respect to (ϕ, λ) . If $v < v'$, v' is called a *proper augmentation*.

Definition 5.8. A Mac Lane pseudovaluation v is called *inductive* if there exists a chain of Mac Lane pseudovaluations

$$v_G = v_0 \leq v_1 \leq \dots \leq v_n = v, \quad (5.8)$$

such that v_i is a proper augmentation of v_{i-1} for $i = 1, \dots, n$. We call $v_0, \dots, v_n$ the *augmentation chain* representing v and write

$$v = [v_0, v_1(\phi_1) = \lambda_1, \dots, v_n(\phi_n) = \lambda_n]. \quad (5.9)$$

Here $v_0 = v_G$, $v_n = v$, the polynomial ϕ_i is a key polynomial for v_{i-1} , $\lambda_i > v_{i-1}(\phi_i)$, and $v_i = [v_{i-1}, v_i(\phi_i) = \lambda_i]$. Also, $\lambda_i \neq \infty$ for $i < n$. We say that the augmentation chain $v_0, \dots, v_n$ is *minimal* if

$$\deg \phi_1 < \dots < \deg \phi_n. \quad (5.10)$$

5.3 Minimal augmentation chains

The central structure result for these constructions describes Mac Lane pseudovaluations by minimal augmentation chains.

The following result combines the existence statements in [Rü14, Theorem 4.31 and Corollary 4.67] with the uniqueness criterion in [Rü14, Theorem 4.33].

Theorem 5.9 (Minimal augmentation chains). (i) Every $v \in V(K[x])$ is inductive and can be represented by a minimal augmentation chain.

(ii) If K is complete, then every $v \in V(K[x])^*$ is inductive and can be represented by a minimal augmentation chain.

(iii) Let

$$\begin{aligned} v_n &= [v_0, v_1(\phi_1) = \lambda_1, \dots, v_n(\phi_n) = \lambda_n], \\ w_m &= [v_0, w_1(\psi_1) = \mu_1, \dots, w_m(\psi_m) = \mu_m] \end{aligned}$$

be inductive pseudovaluations represented by minimal augmentation chains. Then $v_n = w_m$ if and only if $m = n$ and, for every $i = 1, \dots, n$,

- (a) $\lambda_i = \mu_i$,
- (b) $\deg \phi_i = \deg \psi_i$, and
- (c) $v_{i-1}(\phi_i - \psi_i) \geq \lambda_i$.

5.4 Discs, discoids, and valuations

We conclude by introducing discs and discoids in $\overline{K}$, which give a geometric description of subsets defined by valuation conditions on polynomials.

Definition 5.10. (a) Let $\alpha \in \overline{K}$ and let $\lambda \in \hat{\mathbb{Q}}$. Then

$$D(\alpha, \lambda) := \{x \in \overline{K} : v_K(x - \alpha) \geq \lambda\} \quad (5.11)$$

is the (closed) disc with centre α and radius λ .

(b) Let $\phi \in K[x]$ be a monic irreducible polynomial and let $\lambda \in \hat{\mathbb{Q}}$. The set

$$D(\phi, \lambda) := \{x \in \overline{K} : v_K(\phi(x)) \geq \lambda\} \quad (5.12)$$

is called the discoid with centre ϕ and radius λ .

Remark 5.11. For the following description, assume that K is complete, as in the cited source. Any disc with centre in K is a discoid. Conversely, if $\phi = x - \alpha$ then $D(\phi, \lambda)$ coincides with the disc $D(\alpha, \lambda)$. More generally, let $\phi \in K[x]$ be monic and irreducible. Let $\alpha_1, \dots, \alpha_n \in \overline{K}$ be the roots of $\phi \in K[x]$, listed with multiplicity, so that $n = \deg \phi$, and let $\mu_1, \dots, \mu_n \in \hat{\mathbb{Q}}$ be minimal¹ such that $D(\alpha_i, \mu_i) \subset D(\phi, \lambda)$. Then

$$D(\phi, \lambda) = \bigcup_{i=1}^n D(\alpha_i, \mu_i), \quad (5.13)$$

where the discs in this union need not be pairwise distinct; see [Rü14, Lemma 4.43].

¹Our definition of discs implies that discs with large radius are smaller than those with small radius. For instance, the disc with centre $0 \in K$ and radius 0 is relatively large, as it consists of all integral elements.

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